

THE
JUGGLER'S ORACLE;

OR,
THE WHOLE ART
OF
Legerdemain Laid Open.

Illustrated by upwards of Forty Wood Engravings.



LONDON:
PRINTED FOR WILLIAM COLE,
10, NEWGATE STREET.

One Shilling.

THE
JUGGLER'S ORACLE;

OR,
THE WHOLE ART
OF

Legerdemain Laid Open.

Illustrated by upwards of Forty Wood Engravings.



LONDON:
PRINTED FOR WILLIAM COLE,
10, NEWGATE STREET,

One Shilling.

The Project Gutenberg eBook of The Juggler's Oracle; or, The Whole Art of Legerdemain Laid Open

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The Juggler's Oracle; or, The Whole Art of Legerdemain Laid Open

Author: H. Boaz

Release date: July 12, 2018 [eBook #57487]

Language: English

Other information and formats: www.gutenberg.org/ebooks/57487

Credits: Produced by Charlie Howard and the Online Distributed Proofreading Team at <http://www.pgdp.net> (This file was produced from images generously made available by the Library of Congress)

*** START OF THE PROJECT GUTENBERG EBOOK THE JUGGLER'S ORACLE; OR, THE WHOLE ART OF LEGERDEMAIN LAID OPEN

**THE
JUGGLER'S ORACLE;**

OR,
THE WHOLE ART
OF
Legerdemain Laid Open:

CONSISTING OF
ALL THE NEWEST AND MOST SURPRISING
TRICKS AND EXPERIMENTS,

WITH
CARDS,
CUPS AND BALLS,
CONVEYANCE OF MONEY AND RINGS,
BOXES,
FIRE,
STRINGS AND KNOTS;

WITH
MANY CURIOUS EXPERIMENTS

*By Optical Illusion, Chymical Changes, and
Magical Cards, &c.*

THE WHOLE
Illustrated by upwards of Forty Wood Engravings.

BY THE SIEUR H. BOAZ,
THIRTY YEARS PROFESSOR OF THE ART.

LONDON:
PRINTED FOR WILLIAM COLE,
10, NEWGATE STREET.

PRINTED BY G. H. DAVIDSON,
IRELAND YARD, DOCTORS' COMMONS.

CONTENTS.

	Page.
Description of the Operator	<u>1</u>

[TRICKS WITH CARDS.](#)

1. To deliver Four Aces, and to convert them into Knives	<u>2</u>
2. Method of making the Pass	<u>3</u>
3. The Card of Divination	<u>4</u>
Another Way	<u>5</u>
4. The Four Confederate Cards	<u>5</u>
5. The Fifteen Thousand Livres	<u>5</u>
6. The Magic Ring	<u>7</u>
7. The Card in the Mirror	<u>8</u>
8. The Marvellous Vase	<u>9</u>
9. The Nerve Trick	<u>10</u>
10. To make the Constable catch the Knave	<u>11</u>
11. To change a Card into a King or Queen	<u>11</u>
12. To tell a Person what Card he took Notice of	<u>12</u>
13. To tell what Card is at the Bottom, when the Pack is shuffled	<u>12</u>
14. Another Way, not having seen the Cards	<u>12</u>
15. To tell, without Confederacy, what Card one thinks of	<u>13</u>
16. To make a Card jump out of the Pack, and run on the Table	<u>13</u>
17. To tell a Card, and to convey the Same into a Nut or Cherry-Stone	<u>13</u>

18. To let Twenty Gentlemen draw Twenty Cards, and to make one Card every Man's Card	<u>13</u>
19. To transform the Four Kings into Aces, and afterwards to render them all Blank Cards	<u>13</u>
20. To name all the Cards in the Pack, and yet never see them	<u>15</u>
21. To show any one what Card he takes Notice of	<u>15</u>
22. To tell the Number of Spots on the Bottom Cards, laid down in several Heaps	<u>16</u>
23. To make any two Cards come together which may be named	<u>17</u>
24. Card nailed to the Wall by a Pistol-shot	<u>17</u>
25. To tell what Card one thinks of	<u>19</u>
26. Another Way	<u>19</u>
27. To make a Card jump out of an Egg	<u>20</u>
28. The Little Sportsman	<u>20</u>

[CUPS AND BALLS.](#)

29. To pass the Balls through the Cups	<u>22</u>
30. A still more Extraordinary Mode of Playing at Cups and Balls	<u>26</u>

[CONVEYANCE OF MONEY, &c.](#)

31. To convey Money from one Hand to the other	<u>28</u>
32. To convert Money into Counters, and the Reverse	<u>28</u>
33. To put a Sixpence into each Hand, and, with Words, bring them together	<u>29</u>
34. To put a Sixpence into a Stranger's Hand, and another into your own, and to convey both into the Stranger's Hand with Words	<u>29</u>
35. To show the same Feat otherwise	<u>29</u>

36. To throw a Piece of Money away, and find it again	<u>30</u>
37. To make a Sixpence leap out of a Pot or to run along a Table	<u>30</u>
38. To make a Sixpence sink through a Table, and to vanish out of a Handkerchief	<u>30</u>
39. To know if a Coin be a Head or Woman, and the Party to stand in another Room	<u>31</u>
40. To command Seven Halfpence through the Table	<u>31</u>
41. To command a Sixpence out of a Box	<u>32</u>
42. To blow a Sixpence out of another Man's Hand	<u>32</u>
43. To make a Ring shift from one Hand to another, and to make it go on whatever Finger is required, while Somebody holds both Arms	<u>33</u>
44. To transfer a Counter into a Silver Groat	<u>34</u>
45. To make a Silver Twopence be plain in the Palm of your Hand, and be passed from thence wherever you like	<u>35</u>
46. To convey a Sixpence out of the Hand of one that holds it fast	<u>35</u>
47. To convey a Shilling from one Hand into another, holding your Hands apart	<u>36</u>
48. To transform any small Thing into any other Form, by folding of Paper	<u>36</u>
49. Another Trick of the same Nature	<u>36</u>
50. A Watch recovered after being beaten to Pieces in a Mortar	<u>37</u>

[TRICKS WITH BOXES, &c.](#)

51. The Egg-Box	<u>38</u>
52. The Penetrative Guinea	<u>39</u>
53. The Chest which opens at Command	<u>40</u>
54. The Melting-Box	<u>41</u>
55. Trick upon the Globe-Box	<u>42</u>

56. Trick with the Funnel	<u>44</u>
57. The Magical Bell and Bushel	<u>44</u>
58. Out of an Empty Bag to bring upwards of an Hundred Eggs; and, afterwards, a living Fowl	<u>45</u>
59. Bonus Genius; or, Hiccius Doctius	<u>46</u>
60. To make a Knife leap out of a Pot	<u>47</u>
61. To turn a Box of Bird-seed into a living Bird	<u>47</u>

[EXPERIMENTS WITH FIRE.](#)

62. To produce a Carmine Red Flame	<u>48</u>
63. An Orange-coloured Flame	<u>48</u>
64. To make Balloons with Soap and Water that catch Fire and detonate	<u>48</u>
65. A Brilliant Blue Flame	<u>49</u>
66. An Emerald Green Flame	<u>49</u>
67. Loud Detonations, like the Discharge of Artillery	<u>49</u>
68. A Well of Fire	<u>50</u>
69. To make a Room seem all on Fire	<u>50</u>
70. To walk on a Hot Iron Bar, without Danger of Burning	<u>50</u>
71. To eat Fire, and blow it up in your Mouth with a Pair of Bellows	<u>50</u>
72. To Light a Candle by a Glass of Water	<u>52</u>
73. Fulminating Powder	<u>52</u>
74. To set Fire to a Combustible Body by the Reflection of Two Concave Mirrors	<u>52</u>
75. To give the Faces of the Company the Appearance of Death	<u>53</u>
76. To dispose two Little Figures, so that one shall light a Candle, and the other put it out	<u>53</u>
77. To construct a Lantern which will enable a Person to read by Night, at a great Distance	<u>53</u>

TRICKS WITH STRINGS, KNOTS, &c.

- | | |
|--|--------------------|
| 78. To cut a Lace asunder in the Middle, and to make it Whole again | 54 |
| 79. To burn a Thread and make it Whole again with the Ashes | 54 |
| 80. To pull many Yards of Ribbon out of the Mouth | 55 |
| 81. To cut a Piece of Tape into Four Parts, and make it Whole again with Words | 55 |
| 82. To unloose a Knot upon a Handkerchief, by Words | 57 |
| 83. To draw a Cord through the Nose | 58 |
| 84. To take Three Button-Moulds off a String | 59 |

OPTICAL ILLUSIONS.

- | | |
|---|--------------------|
| 85. The Multiplying Mirror | 60 |
| 86. The Magic-Lantern | 61 |
| 87. The Phantascope | 61 |
| 88. The Enchanted Mirrors | 63 |
| 89. The Wonderful Phantoms | 64 |
| 90. The Real Apparition | 65 |
| 91. To draw a Deformed Figure, which will appear well proportioned from a certain Point of View | 67 |

CHEMICAL CHANGES.

- | | |
|--|--------------------|
| 92. To change the Colour of a Rose | 67 |
| 93. To turn Water into Wine | 67 |
| 94. Arbor Dianæ; or, the Silver Tree | 68 |
| 95. The Lead Tree | 68 |
| 96. The Tree of Mars | 68 |
| 97. To form a Metallic Tree, in the Shape of a Fir | 69 |

98. To make a Gold or Silver Tree, to serve as a Chimney Ornament	69
99. Sympathetic or Secret Inks	70
100. Preparation of Green Sympathetic Ink	70
101. Blue Sympathetic Ink	70
102. Yellow Sympathetic Ink	71
103. Purple Sympathetic Ink	71
104. Rose-coloured Sympathetic Ink	71
105. Application of the Secret Inks	71
106. A Drawing which alternately represents Winter and Summer Scenes	71
107. Demonstration of the various Strata of Earth which cover the Globe	72
108. To freeze Water in the Midst of Summer, without the Application of Ice	72

MISCELLANEOUS TRICKS AND EXPERIMENTS.

109. To swallow a long Pudding made of Tin	73
110. An artificial Spider	74
111. To pass a Ring through your Cheek	74
112. To cut a Hole in a Cloak, Scarf, or Handkerchief, and by Words to make it Whole again	75
113. The Dancing Egg	75
114. To make three Figures Dance in a Glass	76
115. To shoot a Swallow, and to bring him to Life again	77
116. Singular Trick with a Fowl	77
117. To put a Lock upon a Man's Mouth	77
118. To thrust a Bodkin into the Forehead, without Hurt	79
119. To thrust a Bodkin through your Tongue	79
120. To appear to cut your Arm off, without any Hurt or Danger	80

121. Tricks with a Cat	<u>80</u>
122. To make a Calf's Head bellow, when served up to Table	<u>81</u>
123. To make a Ball rise above the Water	<u>81</u>
124. Mode of sealing Letters, whereby the Impression cannot be taken	<u>81</u>
125. The Enchanted Egg	<u>82</u>
126. To cut a Man's Head off, and to put the Head into a Platter, a Yard from the Body	<u>82</u>
127. To cause Beer to be wrung out of the Handle of a Knife	<u>83</u>
128. To cut a Glass by Heat	<u>84</u>

**THE
JUGGLER'S ORACLE.**

LEGERDEMAIN, OR SLEIGHT-OF-HAND,

Is an art whereby a person seems to work wonderful, incredible, and almost impossible feats. There is no supernatural or infernal agency in the case; for every trick is performed by nimbleness, agility, and effrontery.

The Operator.

The Operator, or Conjuror, should be a person of bold and undaunted resolution, so as to set a good face upon the matter, in case of the occurrence of any mistake whereby a discovery of the nature of the trick in hand may take place by one of the spectators.

He ought to have a great variety of strange terms and high-sounding words at command, so as to grace his actions, amaze the beholders, and draw their attention from the more minute operations.

He ought likewise to use such gestures of body as may help to draw off the attention of the spectators from a strict scrutiny of his actions.

In showing feats and juggling with cards, the principal point consists in the shuffling them nimbly, and always keeping one card either at the bottom or in some known place of the pack, four or five cards from it; hereby you will seem to work wonders, for it will be easy for you to see one card, which, though you be perceived to do it, will not be suspected, if you shuffle them well afterwards; and this caution I must give you, that, in reserving the bottom card, you must always, whilst you shuffle, keep it a little before or a little behind all the cards lying underneath it, bestowing it either a little beyond its fellows before, right over the fore-finger, or else behind the rest, so as the little finger of the left hand may meet with it, which is the easier, readier, and better way. In the beginning of your shuffling, shuffle as thick as you can, and, in the end, throw upon the pack

the nether card, with so many more, at the least, as you would have preserved for any purpose, a little before or a little behind the rest, provided always that your fore-finger (if the pack lie behind) creep up to meet with the bottom card; and, when you feel it, you may then hold it until you have shuffled over the cards again, still leaving your *kept card* below. Being perfect herein, you may do almost what you like with cards by this means: what pack soever you use, though it consist of eight, twelve, or twenty cards, you may keep them still together unsevered, next to the card, and yet shuffle them often, to satisfy the admiring beholders. As for example, and for brevity sake, to show divers feats under one:—

To deliver Four Aces, and to convert them into Knaves.

Make a pack of these eight cards, viz. four knaves and four aces; and, although the eight cards must be immediately together, yet must each knave and ace be evenly set together, and the same eight cards must lie also in the lowest place of the pack; then shuffle them so always, at the second shuffling; so that, at the end of shuffling the said pack, one ace may lie undermost, or so as you may know where it goeth and lieth always: I say, let your aforesaid pack, with three or four cards more, lie inseparable together, immediately upon and with that ace. Then using some speech or other device, and putting your hands, with the cards, to the edge of the table, to hide the action, let out, privately, a piece of the second card, which is one of the knaves, holding forth the pack in both your hands, and showing to the standers-by the nether card, which is the ace, or kept card, covering also the head or piece of the knave, which is the next card, and, with your fore-finger, draw out the same knave, laying it down on the table; then shuffle them again, keeping your pack whole, and so have your two aces lying together at the bottom. And, to reform that disordered card, and also to grace and countenance that action, take off the uppermost card of the bunch, and thrust it into the midst of the cards, and then take away the nethermost card, which is one of your said aces, and bestow it likewise; then you may begin as before, showing another ace, and instead thereof lay down another knave, and so forth, until, instead of four aces, you have laid down four knaves; the spectators, all this while, thinking that four aces lie

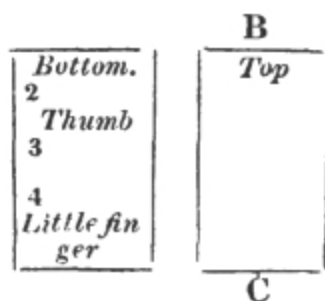
on the table, are greatly amused, and will wonder at the transformation. You must be well practised in shuffling the pack, lest you overshoot yourself.



TRICKS WITH CARDS.

Method of making the Pass.

This art consists in bringing a certain number of cards from the bottom of the pack to the top. I shall explain the method of doing it, before proceeding further, as many of the following recreations depend on the dexterous performance of this manœuvre. Hold the pack of cards in your right hand, so that the palm of your hand may be under the cards. Place the thumb of that hand on one side of the pack, the first, second, and third fingers on the other side, and your little finger between those cards that are to be brought to the top, and the rest of the pack. Then place your left hand over the cards, in such a manner that the thumb may be at C, and the fingers at B, according to the following figures:—



The hands and the two parts of the cards being thus disposed, you draw off the lower cards, confined by the little finger and the other parts of the right hand, and place them, with an imperceptible motion, on the top of the pack. It is necessary, before you attempt any of the recreations that depend on making the pass, that you can perform it so dexterously that the eye cannot distinguish the motion of your hand; otherwise, instead of deceiving

others, you will expose yourself. It is also proper that the cards make no noise, as that will occasion suspicion. This dexterity is not to be attained without some practice. It is sometimes usual to prepare a pack of cards, by inserting one or more that are a small matter longer or wider than the rest; and that preparation will be necessary in some of the following recreations.

The Card of Divination.

Have a pack in which there is a longer card than the rest; open the pack at that part where the long card is, and present the pack to a person in such a manner that he will naturally draw that card. He is then to put it into any part of the pack, and shuffle the cards. You take the pack and offer the same card in like manner to a second or third person; observing, however, that they do not stand near enough to observe the card each other draws. You then draw several cards yourself, among which is the long card; and ask each of the parties if his card be among those cards, and he will naturally say yes, as they have all drawn the same card. You then shuffle all the cards together, and, cutting them at the long card, you hold it before the first person, so that the others may not see it, and tell him that is his card. You then put it again in the pack, and, shuffling them a second time, you cut again at the same card, and hold it in like manner to the second person; and so of the rest.

If the first person should not draw the long card, each of the parties must draw different cards; when, cutting the pack at the long card, you put those they have drawn over it, and, seeming to shuffle the cards indiscriminately, you cut them again at the long card, and show one of them his card. You then shuffle and cut again, in the same manner, and show another person his card, and so on; remembering that the card drawn by the last person is the first next the long card; and so of the others.

Another Way.

This recreation may be performed without the long card, in the following manner: let a person draw any card whatever, and replace it in the pack; you then make the pass, and bring that card to the top of the pack, and shuffle them without losing sight of that card. You then offer that card to a second person, that he may draw it, and put it in the middle of the pack. You make the pass, and shuffle the cards a second time, in the same manner, and offer the card to a third person; and so again to a fourth or fifth.

The Four Confederate Cards.

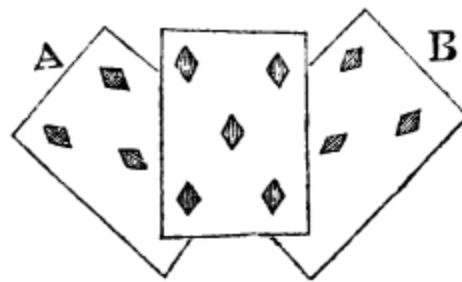
You let a person draw any four cards from the pack, and tell him to think of one of them. When he returns you the four cards, you dexterously place two of them under the pack, and two on the top. Under those at the bottom you place four cards of any sort, and then, taking eight or ten from the bottom cards, you spread them on the table, and ask the person if the card he fixed on be among them. If he say no, you are sure it is one of the two cards on the top. You then pass those two cards to the bottom, and, drawing off the lowest of them, you ask if that is not his card. If he again say no, you take that card up, and bid him draw his card from the bottom of the pack. If the person says his card is among those you first drew from the bottom, you must dexterously take up the four cards that you put under them, and, placing those on the top, let the other two be the bottom cards of the pack, which you are to draw in the manner before described.

The Fifteen Thousand Livres.

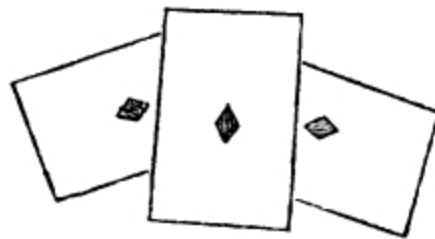
You must be prepared with two cards like the following:—



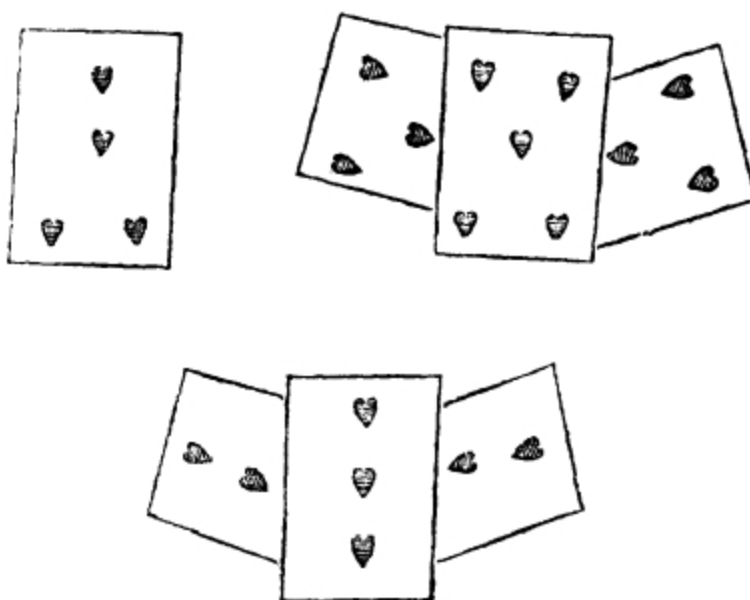
and with a common ace and five of diamonds. The five of diamonds, and the two prepared cards, are to be disposed thus:—



and, holding them in your hand, you say,—“A certain Frenchman left fifteen thousand livres, which are represented by these three cards, to his three sons; the two youngest agreed to leave their 5000, each of them, in the hands of the elder, that he might improve it.” While you are telling this story, you lay the five on the table, and put the ace in its place, and at the same time artfully change the position of the other two cards, that the three cards may appear as in the following figure:—



You then resume your discourse. “The eldest brother, instead of improving the money, lost it all by gaming, except three thousand livres, as you here see.” You then lay the ace on the table, and, taking up the five, continue your story. “The eldest, sorry for having lost the money, went to the East Indies with these 3000, and brought back 15,000.” You then show the cards in the same position as at first. To render this deception agreeable, it must be performed with dexterity, and should not be repeated, but the cards immediately put in the pocket; and you should have five common cards in your pocket, ready to show, if any one should desire to see them. Another recreation of this sort may be performed with fives and threes, as follows:—



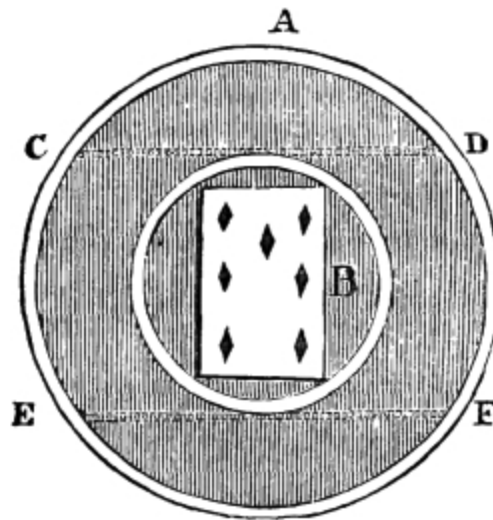
The Magic Ring.

Make a ring large enough to go on the second or third finger, in which let there be set a large transparent stone, to the bottom of which must be fixed a small piece of black silk, that may be either drawn aside or expanded by turning the stone round. Under the silk is to be the figure of a small card. Then make a person draw the same sort of card as that at the bottom of the ring, and tell him to burn it in the candle. Having first shown

him the ring, you take part of the burnt card, and, reducing it to powder, you rub the stone with it; and, at the same time, turn it artfully round, so that the small card at bottom may come in view.

The Card in the Mirror.

Provide a mirror, either round, like the following figure,



or oval; the frame of which must be at least as wide as a card. The glass in the centre must be made to move in the two grooves, C D and E F; and so much of the silvering must be scraped off as is equal to the size of a common card. Observe that the glass be likewise wider than the width of the card. Then paste over the part where the quicksilver is rubbed off a piece of pasteboard, on which affix a card that exactly fits the space, which must at first be placed behind the frame. This mirror must be placed against a partition, through which are to go two strings, by which an assistant in the adjoining room can easily move the glass in the grooves, and consequently make the card appear or disappear at pleasure.

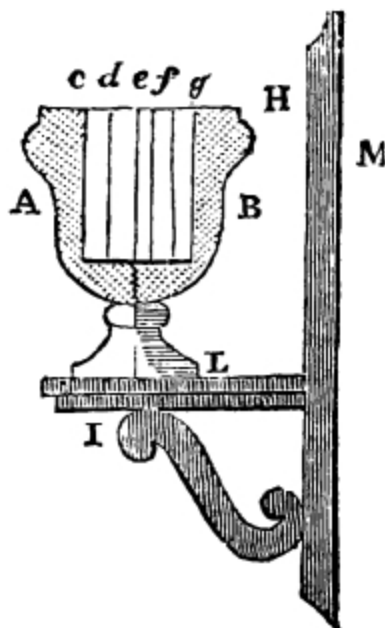
Matters being thus prepared, you contrive to make a person draw the same sort of card with that fixed to the mirror, and place it in the middle of

the pack: you then make the pass, and bring it to the bottom: you then direct the person to look for his card in the mirror, when the confederate, behind the partition, is to draw it slowly forward, and it will appear as if placed between the glass and the quicksilver. While the glass is being drawn forward, you slide off the card from the bottom of the pack, and convey it away. The card fixed to the mirror may easily be changed each time the experiment is performed. This recreation may also be made with a print that has a glass before it, and a frame of sufficient width, by making a slit in the frame, through which the card is to pass; but the effect will not be so striking as in the mirror.

The Marvellous Vase.

Place a vase of wood or pasteboard, like the following:—

A B, on a bracket L, to the partition M.



Let the inside of this vase be divided into five parts, *c, d, e, f, g*; and let the divisions *c* and *d* be wide enough to admit a pack of cards, and those of *e, f, g*, one card only. Fix a thread of silk at the point H, the end of which, passing down the division *d*, and over the pulley I, runs along the bracket L, and goes out behind the partition M. Take three cards from a piquet pack, and place one of them in each of the divisions, *e, f, g*, making the silk thread or line go under each of them. In the division *c* put the pack of cards, from which you have taken the three cards that are in the other divisions. Then take another pack of cards, at the bottom of which are to be three cards of the same sort with those in the three small divisions, and, making the pass, bring them to the middle of the pack, and let them be drawn by three different persons. Then give them all the cards to shuffle, after which place the pack in the division *d*, and tell the parties they shall see the three cards they drew come, at their command, separately out of the vase. An assistant behind the partition then drawing the line with a gentle and equal motion, the three cards will gradually rise out of the vase. Then take the cards out of the division *c*, and show that those three cards are gone from the pack. The vase must be placed so high that the inside cannot be seen by the company. You may perform this recreation also without an assistant, by fixing a weight to the end of the silk line, which is to be placed on a support, and let down at pleasure, by means of a spring in the partition.

The Nerve Trick.

Having previously looked at a card, bid the person draw one, taking care to show him that which you know; when he has it, let him put it at the bottom; let him shuffle the cards, then look at them again, and, finding the card, place it at the bottom; then cut them in half; give the party that half which contains his chosen card at the bottom, to hold between his finger and thumb, just at the corner; bid him pinch them as tight as he can; then striking them pretty sharply, they will all fall to the ground, except the bottom one, which is the card he has chosen. This is a very curious trick, and, if cleanly done, is really astonishing. It may be accounted for from the nature of the nerves, which are always more retentive when any thing is attempted to be taken, either by force or surprise.

To make the Constable catch the Knave.

Take a pack of cards, and look out the four knaves; lay one of them privately on the top of the pack, and the other three down upon the table, saying, “*Here, you see, are three knaves got together, about no good you may be sure;*” then lay down a king beside them, saying, “*But here comes the constable and catches them together: ‘Oh, says he, have I caught you? Well, the next time I catch you together, I’ll punish you severely for all your rogueries.’ ‘Oh,’ but say they, ‘you shan’t catch us again together in haste;’ so they conclude to run three several ways. ‘Well, I’ll go here,’ says one (so take one of the knaves, and put him at the top of the pack). ‘And I’ll go here,’ says another (so put him at the bottom). ‘Then I’ll go here,’ says a third (so put him in the middle). ‘Nay,’ says the constable, ‘if you run, I’ll make sure of one; so I’ll follow the first.’*” Then take the king and put him at the top, and let any one cut the cards asunder two or three times; then deal, cut the cards one by one, and you shall find three together, and the constable with them.

Note.—This feat would be best done with a pack of cards that has two knaves of that sort which you put in the middle.

To change a Card into a King or Queen.

To do this, you must have the picture in your sleeve, and, by a swift sleight, return the card, and fetch out the picture with a back bending. The manner of doing this is better learnt by frequent trials than can be taught by many words. But, if you would do this feat, and yet hold your hand straight and unmoved, then you must peel off the spots or figure of a card, as thin as you can, and just fasten it on the picture with something that will make it stick a little; then, having shown the spots or figure of the card, you may draw it off, and roll it up with your thumb into a very narrow compass, holding it undiscovered, between the inside of the thumb and the ball of your fore-finger; and so produce the picture, to the admiration of the beholders.

To tell a Person what Card he took Notice of.

Take any number of cards, as ten, twelve, and then, holding them with their backs towards you, open four or five of the uppermost, and, as you hold them out to view, let any one note a card, and tell you whether it be the first, second, or third from the top; but you must privately know the whole number of those cards you took. Now shut up your cards in your hands, and take the rest of the pack, which place upon them; then knock their ends and sides upon the table; so it will seem impossible to find the noted card, yet it may easily be done, thus: subtract the number of cards you held in your hand from fifty-two, the whole number in the pack, and to the remainder add the number of the noted card; so the same shall be the number of the noted card from the top; therefore take off the cards one by one, smelling to them till you come to the noted card.

To tell what Card is at the Bottom, when the Pack is shuffled.

When you have seen a card privately, or as though you marked it not, lay the same undermost, and shuffle the cards till your card be again at the bottom. Then show the same to the bystanders, bidding them remember it. Now shuffle the cards, or let any other shuffle them, for you know the card already, and therefore may, at any time, tell them what card they saw, which, nevertheless, must be done with caution or show of difficulty.

Another Way, not having seen the Cards.

If you can see no card, or be suspected to have seen that which you mean to show, then let a bystander shuffle, and afterwards take the cards into your own hands, and, having shown them, and not seen the bottom card, shuffle again, and keep the same cards as before you are taught; and either make shift then to see it when their suspicion is past, which may be done by letting some cards fall, or else lay down all the cards in heaps, remembering where you laid the bottom card; then see how many cards lie in some one heap, and lay the flap where your bottom card is upon that

heap, and all the other heaps upon the same; and so, if there were five cards in the heap whereon you laid your card, then the same must be the sixth card, which now you must throw out, or look upon with suspicion, and tell them the card they saw.

To tell, without Confederacy, what Card one thinks of.

Lay three cards at a distance, and bid a bystander be true, and not waver, but think on one of the three, and by his eye you shall assuredly perceive which he thinks on; and you shall do the like if you cast down a whole pack of cards with the faces upwards, whereof there will be few, or none plainly perceived, and they also court cards; but as you cast them down suddenly, so must you take them up presently, marking both his eyes and the card whereon he looks.

To make a Card jump out of the Pack, and run on the Table.

Take a pack of cards, and let any one draw a card that he likes best, and afterwards take and put it into the pack, but so as you know where to find it at pleasure. Then take a piece of wax, and put it under the thumb-nail of your right hand, and thus fasten a hair to your thumb, and the other end of the hair to the card. Now spread the pack of cards open upon the table, and, making use of any technical words or charms, seem to make it jump on the table.

To tell a Card, and to convey the Same into a Nut, or Cherry-Stone.

Take a nut or cherry-stone, and burn a hole through the side of the top of the shell, and also through the kernel, if you will, with a hot bodkin, or bore it with an awl, and with a needle pull out the kernel, so as the same may be as wide as the hole of the shell; then write the name of the card on a

piece of fine paper, and roll it up hard. Put it into the nut or cherry-stone, and stop the hole up with a little wax, and rub the same over with a little dust, and it will not be perceived; then let some bystander draw a card, saying, "*It is no matter what card you draw,*" and, if your hands so serve you to use the cards well, you shall proffer him, and he shall receive, the same card that you have rolled the name of up in the nut. Now take another nut and fill it up with ink, and stop the hole up with wax, and then give that nut which is filled with ink to some boy to crack, and, when he finds the ink come out of his mouth, it will cause great laughter. Many other feats of this kind may be done, so as to keep the company in good humour.

To let Twenty Gentlemen draw Twenty Cards, and to make one Card every Man's Card.

Take a pack of cards and let any gentleman draw one; then let him put it into the pack again, but be sure where to find it again at pleasure; then shuffle the cards again, and let another gentleman draw a card; but be sure that you let him draw no other than the same card as the other drew, and so do for ten or twelve, or as many cards as you think fit; when you have so done, let another gentleman draw another card, but not the same, and put that card into the pack where you have kept the other one, and shuffle them till you have brought both the cards together; then showing the last card to the company, the other will be considered a more wonderful achievement.

To transform the Four Kings into Aces, and afterwards to render them all Blank Cards.

A clever juggler will take four kings in his hand, and apparently show them to the bystanders, and then, after some words and charms, he will throw them down upon the table, taking one of the kings away, and adding but one other card; then taking them up again, and blowing upon them, he will show you them transformed into blank cards, white on both sides; then throwing them down as before, with their faces downwards, he will take them up again, and, blowing upon them, will show you four aces.

This trick is not inferior to any that is performed with cards, and yet is very pretty.

To perform it, you must have cards made for the purpose, *half cards* we may call them; that is, one half kings and the other half aces; so, laying the aces one over the other, nothing but the king will be seen, and then, turning the kings downwards, the four aces will be seen; but you must have two whole cards, one a king to cover one of the aces, or else it will be perceived, and the other an ace to lay over the kings, when you mean to show the aces; then, when you would make them all blank, lay the cards a little lower, and hide the aces, and they will appear all white. The like you may make of four knaves, putting upon them the four fives; and so of the other cards.

To name all the Cards in the Pack, and yet never see them.

To do this, you must privately drop a little water or beer, about the size of a crown-piece, upon the table before which you sit; then rest your elbows upon the table, so that the cuffs of your sleeves may meet, and your hands stick up to the brims of your hat; in this posture your arms will hide the drop of water from the company; then let any one take the cards and shuffle them, and put them into your hands; also let him set a candle before you, for this trick is best done by candlelight. Then holding the cards in your left hand, above the brim of your hat, close up to your head, so that the light of the candle may shine upon the cards, and holding your head down, so in the drop of water, like a looking-glass, you shall see the shadow of all the cards before you. Now draw the finger of your right hand along upon the card, as if you were feeling it, and then lay it down. Thus you may lay down all the cards in the pack, one by one, naming them before you lay them down, which will seem very strange to the beholders, who will think that you have *felt them* out.

To show any one what Card he takes Notice of.

Let any one take a card out of the pack, and note it; then take part of the pack in your hand, and lay the rest down upon the table, bidding him lay his noted card upon them; then, turning your back towards the company, make as though you were looking over the cards in your hand, and put any card at the fore-side; and, whilst you are doing this privately, wait till the cards are laid out in heaps, to find what the bottom cards are. Bid any one take four cards of the same number, viz. four aces, four deuces, four trèys, and any other number not exceeding ten (for he must not take court cards), and lay them out; then take the remaining cards, if any such there be, and divide their number by four, and the quotient shall be the number of spots of the four cards; if twelve cards remain, then on each bottom card are trèys, and, if there be no remaining cards, then the four bottom cards are aces.

To tell the Number of Spots on the Bottom Cards, laid down in several Heaps.

Bid any one take the whole pack of cards in his hand, and, having shuffled them, let him take off the upper card, and, having taken notice of it, let him lay it down upon the table, with its face downwards, and upon it let him lay so many cards as will make up the number of spots on the noted card, *e. g.* twelve. If the card which the person first took notice of was a king, queen, knave, or a single ten, bid him lay down that card with its face downwards, calling it ten; upon that card let him lay another, calling it eleven, and upon that another, calling it twelve; then bid him take off the next uppermost card, saying, “*What is it?*” Suppose it were a nine, laying it down on another part of the table, and calling it nine, upon it let him lay another, calling it ten; upon it another, calling it eleven; and upon it another, calling it twelve; then let him look on the next uppermost card: and so let him proceed to lay them out in a heap, in all respects as before, till he has laid out the whole pack; but, if there be any cards at the last (I mean if there be not enough to make up the last noted card twelve), bid him give them to you; then, to tell him the number of all the spots contained in all the bottom cards of the heaps, do thus:—from the number of heaps subtract four, and multiply the remainder by fifteen, and to the product add the number of those remaining cards which he gave you, if any did remain; but, if there

were but four heaps, then those remaining cards alone show the number of spots sought.

Note.—You ought not to see the bottom cards of the heaps, nor should you see them laid out, or know the number of cards in each heap; it suffices if you know the number of heaps and the number of the remaining cards, if any such there be: and therefore you may perform this feat as well standing in another room as if you were present.—You must have a whole pack.

To make any two Cards come together which may be named.

When any one has named what two cards he would have come together, take the cards and say, “*Let us see if they are here or not, and, if they are, I’ll put them as far asunder as I can;*” then, having found the two cards proposed, dispose them in the pack, and cause them to come together.

This trick would seem much more strange, if, when you have brought the proposed cards together, by laying them in heaps, you lay the heap wherein the proposed cards are at the bottom of the pack, and then shuffle the cards; cut them asunder somewhere in the middle; thus the proposed cards will be found together in the middle of the pack, which will seem very strange.

Card nailed to the Wall by a Pistol-shot.

A card is desired to be drawn, and the person who chooses, is requested to tear off a corner, and to keep it, that he may know the card; the card so torn is then burnt to cinders, and a pistol discharged with gunpowder, with which the ashes of the card are mixed. Instead of a ball, a nail is put into the barrel, which is marked by some of the company; the pack of cards is then thrown up in the air, the pistol is fired, and the card appears *nailed* against the wall. The bit of the corner which was torn off is then compared with it, and is found exactly to fit; and the nail which fastens it to the wall is recognised by the persons who marked it.

Explanation.—When the performer sees that a corner of the chosen card has been torn off, he retires, and makes a similar tear on a like card. Returning, he asks for the chosen card, and passing it to the bottom of the pack, he substitutes expertly in its place the card which he has prepared, which he burns instead of the first. When the pistol is loaded, he takes it in his hand, under the pretence of showing how to direct it, &c. He avails himself of this opportunity to open a hole in the barrel, near the touchhole, through which the nail falls by its own weight into his hand: having shut this passage, he requests one of the company to put more powder and wadding into the pistol; whilst that is doing, he carries the nail and card to his confederate, who quickly nails the card to a piece of square wood, which stops a space left open in the partition, but which is not perceived, as it is covered with a piece of tapestry, similar to the rest of the room; and by which means, when the nailed card is put on, it is not perceived; the piece of tapestry which covers it is nicely fastened on the one end by two pins, and on the other a thread is fastened, one end of which the confederate holds in his hand. As soon as the report of the pistol is heard, the confederate draws his thread, by which means the piece of tapestry falls behind a glass; the card appears the same that was marked, and with the nail that was put in the pistol. It is not astonishing that the trick, being so difficult by its complexity to be guessed at, should have received such universal applause.

N. B. After the pistol has been charged with powder, a tin tube may be slipped upon the charge, into which the nail being rammed along with the wadding, by inclining it a little, in presenting it to one of the spectators to fire, the tube and contents will fall into the performer's hand, to convey to his confederate. If any one suspects that the nail has been stolen out of the pistol, you persist in the contrary, and beg the company at the next exhibition to be farther convinced; you then are to show a pistol, which you take to pieces, to show that all is fair, without any preparation; you charge it with a nail, which is marked by some person in confederacy with you, or you show it to many people, on purpose to avoid its being marked. In this case the card is nailed with another nail; but, to persuade the company that it is the same, you boldly assert that the nail was marked by several persons, and you request the spectators to view it, and be convinced.

To tell what Card one thinks on.

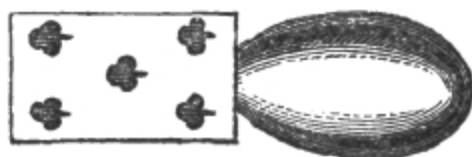
Take twenty-one cards, and begin to lay them down three in a row, with their faces upwards; then begin again at the left hand, and lay one card upon the first, and so on the right hand, and then begin at the left hand again, and so go on to the right; do this till you have laid out the twenty-one cards in three heaps, but, as you are laying them out, bid any one think of a card, and, when you have laid them all out, ask him which heap his card is in; then lay that heap in the middle, between the other two. Now lay them all out again into three heaps, as before, and, as you lay them out, bid him take notice where his noted card goes, and put that heap in the middle, as before. Then, taking the cards with their backs towards you, take off the uppermost card, and, smelling to it, reckon it one; then take off another, and, smelling to it, reckon it two; this do till you come to the eleventh card, for that will always be the noted card after the third time of laying them out, though you should lay in this manner ever so often. You must never lay out the cards less than three times, but as often above as you please. This trick may be done by any odd number of cards that may be divided by three.

Another Way to tell what Card is noted.

When one has noted a card, take it and put it at the bottom of the pack; then shuffle the cards till it comes again to the bottom; then see what is the noted card, which you may do without being taken notice of; when you have thus shuffled the cards, turn them with their faces towards you, and knock their ends upon the table, as though you would knock them level; and, whilst you are so doing, take notice of the bottom card, which you may do without suspicion, especially having shuffled them before; then, when you know the card, shuffle them again, and give them to any of the company, and let them shuffle them, for you know the card already, and may easily find it at any time.

To make a Card jump out of an Egg.

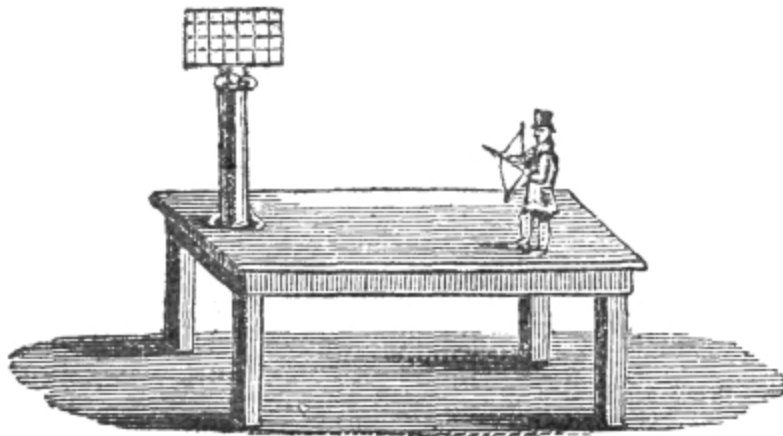
To do this wonderful feat you must have two sticks, made both of one size, so that no person can know one from another; one of these sticks must be made so as to conceal a card in the middle, thus: you must have one of your sticks turned hollow quite through, and then an artificial spring to throw the card into the egg at pleasure. The operation is thus:—Take and peel any card in the pack, which you please, and roll it up. Now put it into the false stick, and there let it lie until you have occasion to make use of it. Now take a pack of cards, and let any body draw one; but be sure to let it be the same sort of card which you have in the hollow stick. The person who has chosen it is now to put it into the pack again, and, while you are shuffling them, let it fall into your lap. Then, calling for some eggs, desire the person who drew the card, or any other person in the company, to choose any one of the eggs. When they have done so, ask the person if there be any thing in it? He will answer that there is not. Now take the egg in your left hand, and the hollow stick in your right, and then break the egg with the stick.



Having let the spring go, the card will appear in the egg, to the great amazement of the beholders. Now be sure to conceal the hollow stick, and produce the solid one, which place upon the table for the examination of those who are curious.

The Little Sportsman.

This is a little pasteboard figure which holds a bow, with an arrow, which it shoots at the instant required, and hits a paper placed opposite, on the top of a pedestal:



This paper is divided into several squares, which are numbered, and the arrow flies and always hits the number chosen by one of the company. The action of the spring which impels it is restrained by a little pin, which the confederate lets go at pleasure, by moving the levers hid in the table; when you push this pin, the arrow flies with rapidity to the paper, like the operation of a lock of a musket, when you pull the trigger. In placing the automaton on the table, you may place it in such a manner that the arrow be directed towards one of the circles numbered on the paper. To cause that number to be chosen against which the arrow is pointed, you must present to the spectator cards numbered, and dexterously make him choose the number required, which depends on peculiar address, that is scarcely possible to be described by words; it may in general be said to come under the following heads:—First, to put at the bottom one of the cards to be chosen; secondly, to keep it always in the same place, although you mix, or pretend to mix, the cards; thirdly, to pass the card to the middle, when you present the pack; fourthly, to pass many cards before the hands of the spectator, to persuade him that he may choose indifferently; fifthly, to pass these same cards with such rapidity, that he cannot take any but the card intended; sixthly, to slip complaisantly into his hand the card you wish to be taken, at the very moment when, the better to deceive him, you beg of him graciously to take which card he chooses.

CUPS AND BALLS.

To pass the Balls through the Cups.

You must place yourself at the farther end of the table, and provide three cups made of tin; you must likewise have your black magical stick, to show your wonders withal. You are also to provide four small cork balls, to play with; but do not let any more than three of them be seen upon the table.

N. B.—Always conceal one ball in the right hand, between the middle finger and the ring finger; and be sure you make yourself perfect to hold it there, for by this means all the tricks of the cups are done.

Now say something to the following effect:—

Three cups, you see, have I got here,
'Tis true they are but tin;
Silver and gold are much too dear
For me to conjure in.

Now, Ladies and Gentlemen, here are no equivocations at all;
But, if your eyes are not as quick as my hands, I shall deceive you all.

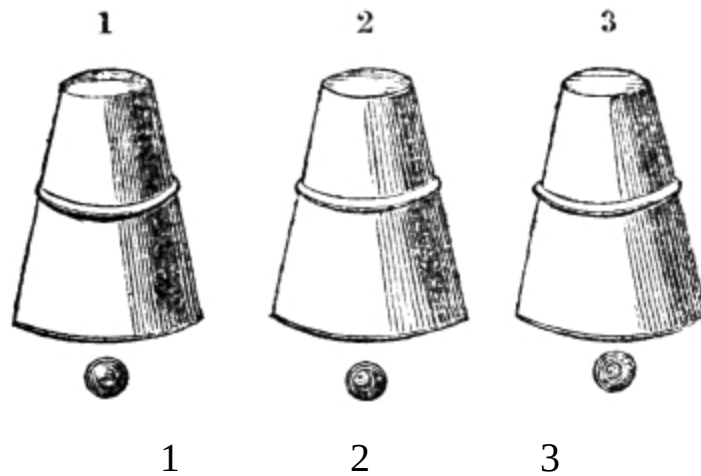
View them well within,
View them all round about;
Where there's nothing in,
There's nothing can come out.

Then take your balls privately between your fingers, fling one of them upon the table, and say thus—

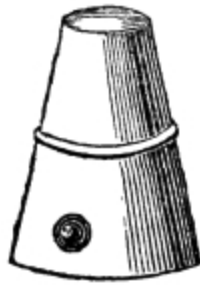
The first trick that I learned to do
Was out of one ball to make two:
Ah! since it cannot better be,
Of these two I will soon make three,
Which is called the first trick of dexterity.

So then you have three balls on the table to play with, and one left between the fingers of your right hand.

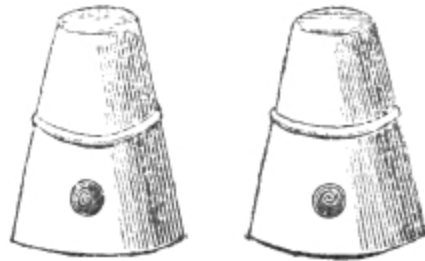
The Position of the Cups is thus—



Lay your three balls upon the table; then say, “*Ladies and Gentlemen, you see here are three balls, and here are three cups; that is, a cup for each ball, and a ball for each cup.*” Then taking the ball which you have in your right hand (which you are always to keep private) and clapping it under the first cup; and, taking up one of the three balls with your right hand, seem to put it into your left hand, but retain it still in your right, shutting your left hand in due time; then say, “*Presto, begone.*”



Then taking the second cup up, say, "*Ladies and Gentlemen, you see there is nothing under my cup;*" so clap the ball under that you have in your right hand, and then take the second ball up with your right hand, and seem to put it into your left, but retain it in your right, shutting your left hand in due time, as before, saying, "*Vado, begone.*"



Then taking the third cup up, saying, "*Ladies and Gentlemen, you see there is nothing under my last cup;*" clap the ball under your right hand, and, taking the third ball up with your right hand, seem to put it into your left hand, but retain it in your right hand; so, shutting your left hand in due time, as before, saying, "*Presto, make haste*": so you have your three balls come under your three cups, as thus, and so lay your three cups down upon the table.



Then with your right hand take up the first cup, and clap the ball under that you have in your right hand; saying, “*Ladies and Gentlemen, this being the first ball, I’ll put it in my pocket;*” but that you must still keep in your right hand to play withal.



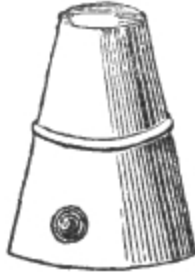
So take up the second cup with your right hand, and clap that ball under which you have concealed, and then take up the second ball with your right hand, and say, “*This likewise I take and put into my pocket.*”

Likewise take up the third cup, and, clapping the cup down again, convey the ball that you have in your right hand under the cup; then take the third ball, and say, “*Ladies and Gentlemen, this being the last ball, I take this and put it in my pocket.*” Likewise then say to the company, “*Ladies and Gentlemen, by a little of my fine powder of experience I’ll command these balls under the cups again,*” as thus:—



So lay them all along the table, to the admiration of the beholders.

Then take up the first cup, and, clapping the ball under that you have in your right hand, and taking the first ball up with the right hand, seem to put the same into your left hand, but retain it still in your right hand; then say, "*Vado, quick, begone when I bid you, and run under the cup.*"



Then taking that cup up again, and flinging it under that you have in your right hand, you must take up the second ball, and seem to put it into your left hand, but retain it in your right, saying, "*Ladies and Gentlemen, see how the ball runs on the table.*"—So, seeming to fling it away, it will appear thus:—



Now, taking the same cup up again, clap the ball under as before; and, taking the third ball in your right hand, seem to put it under your left, but still retain it in your right hand; then with your left hand seem to fling it in the cup, and it will appear thus:—



All the balls being under one cup.

If you can perform these feats with the cups and balls dexterously, you may change the balls into apples, pears, plums, or living birds, as your fancy leads you.

A still more Extraordinary Mode of Playing at Cups and Balls.

You must provide six cups made of the same size and metal (persons with hands, as seen above, require only three), but keep three of them concealed in the juggling-bag until they are required.

The trick performed by the three first cups is as follows:—Take out of the bag your three cups, and place them on a table. You must have balls of cork provided, and concealed, but one ball must be on the table. Then say, “*Ladies and Gentlemen,*” turning up your three cups (though at the same time you must have a ball concealed), “*you see there is nothing under my cups; I take and put this cup here; I put the second there; and the third there.*” The ball you have hid must be clapped under one of the cups at the time you were placing them.

You must have a tin bottom in the inside of one of your cups, and holes punched in it like a grater. Then say, “*Ladies and Gentlemen,*” (taking the ball off the table, and placing it on the cup the ball is under), “*observe, I cover this ball with this cup,*” clapping the third cup on the other two; then say, “*Presto, I command the ball from under the middle cup to the bottom.*” Then taking off the first and second cups, the ball, they think, is gone to the bottom; whereas the ball that is laid on the top of the undermost sticks fast to the grater that covers it, and, when the cup is turned up, it is the ball that was conveyed first that appears.

Next place, take the ball that is on the table, and say, "*Ladies and Gentlemen, there is but one ball left,*" clapping the cup with the tin bottom, where the ball is concealed, over that ball on the table, so as that ball that was sticking to the tin falls down, and makes two; then clapping the cup down, convey another ball you have secured, and say, "*Vene tome;*" then say, "*Ladies and Gentlemen, there are three balls and three cups;*" first having secured two balls (having some strange gestures of body and speech to take off the eyes of the spectators), at the same time taking up one of the cups, but putting it down on one of the balls, with the two balls secured; then clap the second cup on the second ball, and the third cup on the third ball.

Now say, "*I've covered the three balls.*" Then turn up one cup, and say, "*There is the first ball;*" then turn up the other, saying, "*There is the second ball.*" Then take up either of the balls, and lay it on the top of the third cup, and cover it with the cup that has the tin bottom, clapping the third cup in the place of the other two; say, "*Ladies and Gentlemen, there is one ball at the bottom, one in the middle, and the third and last ball I strike through the board, saying, 'Presto, be gone.'*"

Now, you must understand, the third ball you drop, the second sticks to the tin grater, and three balls appear under the lowermost cup; then place your three balls on the table, and your cups opposite the balls; then say, "*I cover this ball with this cup, and I cover this third ball with this one cup.*" Then turning up one cup, take up the ball, and say, "*Presto, I command you under the second cup;*" but at the same time you must retain the ball, for the ball that was sticking to the tin is dropped, and makes two; then clapping the cup down, with the ball that you have retained, turn up the cup, and say, "*I'll strike this ball to the other two;*" and drop that ball, being three before.

Next place the balls and cups as they were before; then clap the first cup on the first ball, and the second on the second; then take up the cup with the grater, which generally is in the middle, saying, "*I'll put this cup in my bag;*" and take up this ball, saying, "*I'll put this ball in my bag;*" and take up the next ball, saying, "*I'll put this ball in my bag too,*" clapping under the cup at the same time the ball you have retained. At last, say, "*I shall have too many balls,*" or something to that purpose; seem in a fury, and toss your cups away; then put them into your juggling-bag, that, when you show the other three, the company may think they were the first cups.

CONVEYANCE OF MONEY, &c.

The conveyance of money is not much inferior to the tricks with Cups and Balls, but much easier to perform. The principal place to hold the coin is the palm of the hand; and the best piece to play with is a sixpence. But, by practice, all coins will be alike, unless they are very small, and then they must be kept between the fingers, almost at the fingers' ends; whereas the ball is to be kept below, near the palm. The coin ought never to be too large, as that will considerably impede the clever conveyance of it.

To convey Money from one Hand to the Other.

Hold open your right hand and lay therein a sixpence, and on the top of it place the top of your left middle finger, which press hard upon it, at the same time using hard words. Then suddenly draw away your right hand from the left, seeming to have left the coin there, and shut your hand cleverly, as if it still were there. That this may appear to have been truly done, take a knife, and seem to knock against it, so as to make a great sound. This is a pretty trick, and, if well managed, both the eye and ear are deceived at the same time.

To convert Money into Counters, and the Reverse.

Another way to deceive the lookers-on is to do as before, with a sixpence, and, keeping a counter in the palm of your left hand, secretly, seem to put the sixpence therein, which being retained still in the right hand, when the left hand is opened, the sixpence will seem to be turned into a counter.

To put a Sixpence into each Hand, and, with Words, bring them together.

He that hath once attained to the faculty of retaining one piece of money in his right hand may show a hundred pleasant deceits by that means, and may manage two or three as well as one. Thus, you may seem to put one piece into your left hand, and, retaining it still in your right, you may, together therewith, take up another like piece, and so, with words, seem to bring both pieces together. A great variety of tricks may be shown in juggling with money.

To put a Sixpence into a Stranger's Hand, and another into your own, and to convey both into the Stranger's Hand with Words.

You take two sixpences evenly set together, and put the same, instead of one sixpence, into a stranger's hand, and then, making as though you put one sixpence into your own hand, with words, you make it seem that you convey the sixpence in your own into the stranger's hand; for, when you open your said left hand, there shall be nothing seen, and he, opening his hand, shall find two sixpences, which he thought was but one.

To show the same Feat otherwise.

To keep a sixpence between your fingers serves especially for this and such like purposes: hold your hand, and cause one to lay a sixpence upon the palm thereof, then shake the same up almost to your finger's end, and putting your thumb upon it, you may easily, with a little practice, convey the edge betwixt the middle and fore-finger whilst you proffer to put it into your other hand (provided always that the edge appears not through the fingers, on the back side); take up another sixpence, which you may cause another stander-by to lay down, and put them both together, either closely, instead of one into a stranger's hand, or keep them still in your own hand, and, after some words spoken, open your hands, and, there being nothing in

one hand, and both pieces in the other, the beholders will wonder how they came together.

To throw a Piece of Money away, and find it again.

You may, with the middle or ring finger of the right hand, convey a sixpence into the palm, with the same hand, and, seeming to cast it away, keep it still, which, with confederacy, will seem strange: to wit, when you find it again, where another has placed the like piece. But these things cannot be done without practice; therefore. I will proceed to show how things may be brought to pass with less difficulty, and yet as strange as the rest, which, being unknown, are much commended; but, being known, are derided, and nothing at all regarded.

To make a Sixpence leap out of a Pot or to run along a Table.

A juggler takes a sixpence and throws it into a pot, or lays it on the middle of a table, and, with enchanted words, causes the same to leap out of a pot, or run towards him or from him along the table, which seems miraculous till you know how it is done, which is thus: Take a long black hair of a woman's head, fasten it to the rim of a sixpence, by means of a little hole driven through the same with a Spanish needle. In like sort you may use a knife or any small thing; but if you would have it go from you, you must have a confederate, by which means all juggling is graced and amended.

This feat is the stranger if it be done by night, a candle being placed between the spectators and the juggler, for by that means their eyes are hindered from discerning the deceit.

To make a Sixpence sink through a Table, and to vanish out of a Handkerchief.

A juggler will sometimes borrow a sixpence and mark it before you, and seem to put the same in the middle of a white handkerchief, and wind it so as you may the better see and feel it; then he will take the handkerchief and bid you feel whether the sixpence be there or no; and he will also require you to put the same under a candlestick, or some such like thing; then he will send for a basin of water, and, holding the same under the table, right against the candlestick, he will use certain words of enchantment, and, in short, you shall hear the sixpence fall into the basin. This done, let one take off the candlestick, and the juggler take the handkerchief by a tassel and shake it; but the money is gone, which seems as strange a feat as any whatsoever, but, being known, the miracle is turned to a jest; for it is nothing else but to sew a sixpence into a corner of the handkerchief, finely covered with a piece of linen a little bigger than your sixpence, which corner you must convey, instead of the sixpence given you, into the middle of your handkerchief, leaving the other in your hand or lap, which afterwards you seem to pull through the table, letting it fall into the basin.

To know if a Coin be a Head or Woman, and the Party to stand in another Room.

This is done by confederacy: he that lays it down, says, “*What is it?*” and that is a sign it is a head; or he says, “*What is it now?*” and that is a sign it is a woman: cross and pile in silver is done the same way. By confederacy, divers strange things are done: thus, you may throw a piece of money into a pond, and bid a boy go to such a secret place where you have hid it, and he will bring it, and make them believe it is the same that you threw into the pond, and no other.

So let a confederate take a shilling and put it under a candlestick on a table a good distance from you; then you must say, “*Gentlemen, you see this shilling;*” then take your hand and knock it under the table, and convey it into your pocket; then say, “*The shilling is gone; but look under such a candlestick and you will find it.*”

To command Seven Halfpence through the Table.

To do this, you must employ a tinman to make holes, with room enough for a die to go in and out, and let him clap a good halfpenny upon them all, and so make them fast, that nobody can tell them from true ones. Then get a cap to cover your halfpence, also a cap and a die for the company to fling, to amuse them; when you are thus provided, the manner of performing is thus:—Desire any body in the company to lend you seven halfpence, telling them that they will soon be returned; then say, *“Gentlemen, this is made just fit for your money;”* then clapping your cap on, desire somebody in the company to fling that die, and, in so doing, take off the cap, and convey your false money into it, so that the company may not see you put it in; then with your cap cover the die, while with your right hand you take up the true money, and put it into the left under the table, saying, *“Vado, begone; I command the die to be gone, and the money to come in the place;”* so take up the cap, and the die is gone, and the money is come. Having covered the money again with the cap, taking the true money with your right hand, and knocking under the table, also making a jingling, as though the money was coming through the table, fling them on the table, and say, *“There is the money,”* and, with your right hand, take off the cap, saying, *“And there is the die;”* so convey the false money into your lap, and there is the cap likewise.

To command a Sixpence out of a Box.

You must get a box turned with two lids (one must be a false one), and there put the counter, so that it may rattle; and you must have a small peg or button to your box, to hinder the counter from jingling, and at the bottom of the box you must have half a notch made, just fit for a sixpence to come out. So, to perform this feat, you must desire any body to lend you a sixpence, and to mark it with whatever mark he may please; then let him put it into the box himself; afterwards put the cover on, and, by shaking the box, the sixpence will come into your hand, when you may dispose of it as you please.

To blow a Sixpence out of another Man's Hand.

Blow on a sixpence, and immediately clap it into one of the spectator's hands, telling him to hold it fast; then ask him if he is sure he has it. He, to be certain, will open his hand and look. Then say to him, "*Nay, but if you let my breath go off, I cannot do it.*" Then take it out of his hand again, blow on it, and, staring him in the face, clap a piece of horn in his hand, and retain the sixpence, shutting his hand yourself. Bid him hold his hand down, and slip the sixpence into one of his cuffs; then say, "*I command the money you hold in your hand to vanish; Vade, now see.*" When they have looked, they will think it is changed by virtue of your stone. Then take the horn again, and say, "*Vade;*" and then say, "*You have your money again.*" He then will begin to marvel, and say, "*I have not.*" Then say to him again, "*You have; and I am sure you have got it: is it not in your hand? If it be not there, turn down one of your sleeves, for it is in one, I am sure;*" where he, finding it, will not a little wonder.

To make a Ring shift from one Hand to another, and to make it go on whatever Finger is required, while Somebody holds both Arms.

Desire some person in company to lend you a gold ring, recommending him at the same time to make a mark on it, that he may know it again. Have a gold ring of your own, which you are to fasten by a small piece of catgut string to a watch-barrel, which must be sewn to the left sleeve of your coat. Take in your right hand the ring that will be given you: then, taking with dexterity, near the entrance of your sleeve, the other ring fastened to the watch-barrel, draw it to the fingers' ends of your left hand, taking care nobody perceives it. During this operation, hide between the fingers of your right hand the ring that has been lent to you, and fasten it dexterously on a little hook, sewed on purpose on your waistcoat, near your hips, and hid by your coat.

You will, after that, show your ring, which you hold in your left hand; then ask the company on which finger of the other hand they wish it to pass. During this interval, and as soon as the answer has been given, put the

before-mentioned finger on the little hook, in order to slip on it the ring; at that moment let go the other ring, by opening your fingers. The spring which is in the watch-barrel, not being confined longer, will contract and make the ring slip under the sleeve without any body perceiving it, not even those who hold your arms: as, their only attention being to prevent your hands from communicating, they will let you make the necessary motions. These must be very quick, and always accompanied by stamping of the foot.

After this operation, show the assembly that the ring is come on the other hand; make them remark that it is the same that had been lent to you, or that the mark is right. Much dexterity must be made use of to succeed in this entertaining trick, that the deception may not be suspected.

To Transfer a Counter into a Silver Groat.

Take a groat, or a smaller piece of money, and grind it very thin on one side; then take two counters, and grind them, the one on one side, and the other on the other side; glue the smooth side of the groat to the smooth side of the counter, joining them as close together as possible, especially at the edges, which may be so filed that they shall seem to be but one piece; to wit, one side a counter and the other side a groat. Then take a little green wax, for that is softest, and therefore best, and lay it on the smooth side of the counter, as it does not much discolour the groat; and so will that counter, with the groat, cleave together as though they were glued, and, being filed even with the groat and the other counter, it will seem so perfectly like an entire counter, that, though a stranger handle it, he cannot betray it; then, having a little touched your fore-finger, and the thumb of your right hand, with soft wax, take therewith this counterfeit counter, and lay it openly upon the palm of your left hand, wringing the same hard, so as you may leave the glued counter with the groat apparently in the palm of your left hand, and the smooth side of the waxed counter will stick fast upon your thumb, by reason of the wax wherewith it is smeared: and so you may hide it at your pleasure (always be sure to lay the waxed side downward, and the glued side upward); then close your hand, and, in or after closing thereof, turn the piece, and so, instead of a counter, which they suppose to be in

your hand, you shall seem to have a groat, to the astonishment of the beholders, if it be well handled. The juggler must not leave any of his tricks wanting for hard and break-jaw words.

***To make a Silver Twopence be plain in the Palm of your Hand,
and be passed from thence wherever you like.***

Put a little red wax, not too much, upon the nail of your longest finger; then let a stranger put a two-penny piece into the palm of your hand, and shut your fist suddenly, and convey the two-penny piece upon the wax, which, with use, you may so accomplish, as no man shall perceive it; then, and in the meantime, use words of course, and suddenly open your hand; hold the tips of your fingers rather lower than the palm of your hand, and the beholders will wonder where it is gone; then shut your hand suddenly again, and lay a wager whether it be there or no, and you may either leave it there or take it away at pleasure. This, if it be well handled, hath more admiration than any other feat of the hand.—Note: This may be best done by putting the wax upon the two-penny piece, but then you must put it into your hand yourself.

To convey a Sixpence out of the Hand of one that holds it fast.

Stick a little wax upon your thumb, and take a bystander by the fingers, showing him the sixpence, and telling him you will put the same into his hand; then wring it down hard with your waxed thumb, and, using many words, look him in the face, and, as soon as you perceive him look in your face, or on your hand, suddenly take away your thumb, and close his hand, and it will seem to him that the sixpence remains. If you wring a sixpence upon one's forehead, it will seem to stick when it is taken away, especially if it be wet; then cause him to hold his hand still, and, with speed, put out into another man's hand, or into your own, two sixpences instead of one, and use words of course, whereby you shall make the spectators believe, when they open their hands, that, by enchantment, you have brought both together.

To convey a Shilling from one Hand into another, holding your Hands apart.

It is necessary to mingle some merry pranks among your grave miracles, as, in this case of money, to take a shilling in each hand, and, holding your arms abroad, to lay a wager that you will bring them both into one hand without bringing them any nearer together; the wager being laid, hold your arms abroad, like a rod, and, turning about with your body, lay the shilling out of one of your hands, upon the table, and, turning to the other hand, so you shall win your wager.

To transform any small Thing into any other Form, by holding of Paper.

Take a sheet of paper, and fold or double the same, so as one side be a little longer than the other; then put a counter between the two sides of the leaves of the paper, up to the middle of the top of the fold; hold the same so as it be not perceived, and lay a sixpence on the outside thereof, right against the counter, and fold it down to the end of the longer side. When you have unfolded it again, the sixpence will be where the counter was, so that some will suppose you have transformed the money into a counter; and with this many tricks may be done.

Another Trick of the same Nature.

Take two papers, three inches square each, divided into two folds, of three equal parts on either side, so as each folded paper remains one inch square; then glue the back side of the two together, as they are folded, and not as they are opened, and so shall both papers seem to be but one, and, which side soever you open, it shall appear to be the same, if you have handsomely done the bottom, as you may well do with your middle finger, so that, if you have a sixpence in one hand, and a counter in the other, you show but one, and you may, by turning the paper, seem to change it; this is

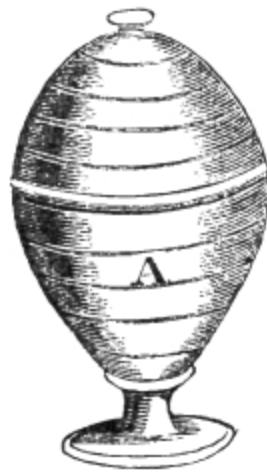
best performed by putting it under a candlestick or a hat, and, with words, seem to do the feat, which is by no means an inferior one.

A Watch recovered after being beaten to Pieces in a Mortar.

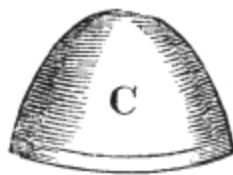
A watch is borrowed from one of the company, and, being put into a mortar, another person is shortly after requested to beat it to pieces with a pestle. It is then shown to the company, entirely bruised; in a few minutes the watch is restored entire to its owner, who acknowledges it to be his property. It is easy to devise that, to effect this, the mortar must be placed near a concerted trap, and that it must be covered with a napkin, to afford an opportunity to the confederate to substitute another watch, unperceived by the company. In order to succeed in the illusion of this trick, you must take care to provide yourself with a second watch, somewhat resembling the first in the size, case, &c. which will not be very difficult, as you may either be furnished with a watch by a person with whom the matter is preconcerted, or by addressing yourself to some one whose watch you have before observed, and procured yourself one like it. After having placed all the pieces in the mortar, you must cover them a second time with a napkin, and whilst you amuse the company with some trick or story, you afford time to your confederate to take the bruised pieces away, and replace the first watch in the mortar.

TRICKS WITH BOXES, &c.

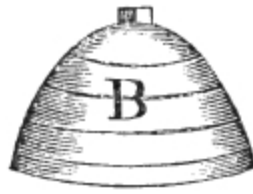
The Egg-Box.



This is the Egg-box, put together like two beehives—one on the top of another. The following is the under-shell,



covered over artificially with the white thin skin of a real egg. The upper shell is of the same shape, but



larger, and is merely the lid of the box. The following



is the lower part of the box. Put B, which is the outward shell, upon C, and both upon D, which arrangement puts all in readiness for the performance of the trick. Now call for an egg, and bid all the bystanders look at it, to see that it is a real egg. Then take off the upper part, B C, with your fore-finger and thumb, and placing the egg in the box, say, "*Ladies and Gentlemen, you see it fairly in the box;*" and, uncovering it again, say, "*You shall see me fairly take it out;*" putting it into your pocket in their sight. Now open your box again, and say, "*There's nothing;*" close your hand about the middle of the box, and, taking B by the bottom, say, "*There is the egg again;*" which appears to the spectators to be the same that you put in your pocket; then clapping that on again, and taking the lid of C between your fore-finger and thumb, say, "*It is gone again.*"

The Penetrative Guinea.

Provide a round tin box, of the size of a large snuff-box, and likewise eight other boxes, which will go easily into each other, and let the least of them be of a size to hold a guinea. Each of these boxes should shut with a hinge, and to the least of them there must be a small lock, that is fastened with a spring, but cannot be opened without a key; and observe, that all these boxes must shut so freely, that they may be all closed at once. Place these boxes in each other, with their tops open, in the drawer of the table on which you make your experiments; or, if you please, in your pocket, in such a manner that they cannot be displaced. Then ask a person to lend you a new guinea, and desire him to mark it, that it may not be changed. You take this piece in one hand, and in the other you have another of the same appearance, and, putting your hand in the drawer, you slip the piece that is marked into the least box, and, shutting them all at once, you take them out. Then showing the piece you have in your hand, and which the company suppose to be the same that was marked, you pretend to make it pass through the box, and dexterously convey it away. You then present the box, for the spectators do not yet know there are more than one, to any person in company, who, when he opens it, finds another, and another, till he comes to the last, but that he cannot open without the key, which you then give him; and, retiring to a distant part of the room, you tell him to take out the guinea himself, and see if it be that he marked.

This trick may be made more surprising by putting the key into the snuff-box of one of the company; which you may do by asking for a pinch of snuff: the key, being very small, will lie concealed among the snuff. When the person who opens the box asks for the key, tell him that one of his friends has it in his snuff-box, which will cause much amazement and merriment. This part of the trick may be done with a confederate.

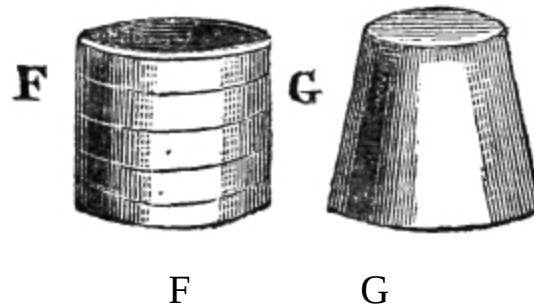
The Chest which opens at Command.

There is a little figure of Mahomet within the chest, in the body of which is a spring, made of brass wire, twisted in a spiral form. By this means the little figure, though higher than the chest, can, by the accommodation of the spring, be contained within when it is shut, as the spring in the body closes and shortens. The chest is placed on levers

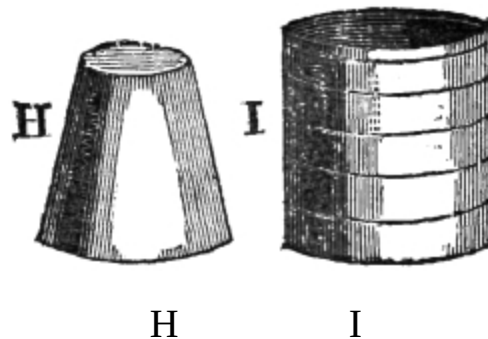
concealed in the table, which communicate their motions by the assistance of the confederate to the bolt and lock. As soon as the staple is disengaged, the spring in the body of the figure, finding no resistance but the weight of the lid, forces it open.

The Melting-Box.

The melting-box is made in the fashion of a screw, so that the lips may hang without discovery.



F is the outer part of the box; G, the first inside part;



H, the second inside part; and I, the round case, made of leather, with a button on the top, and wide enough to slip on and off, half in the bottom of the box, into which put a small quantity of quicksilver, killed or

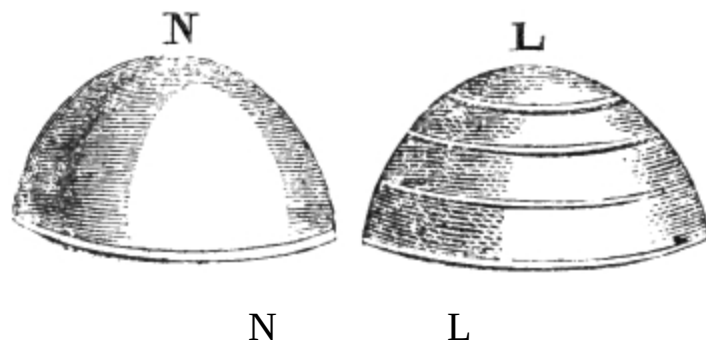
amalgamated, which may be done with the shavings of pewter. In the second part, which is H, let there be six single pence: put these in the first or outermost part, then put G to H, and the box is perfect.

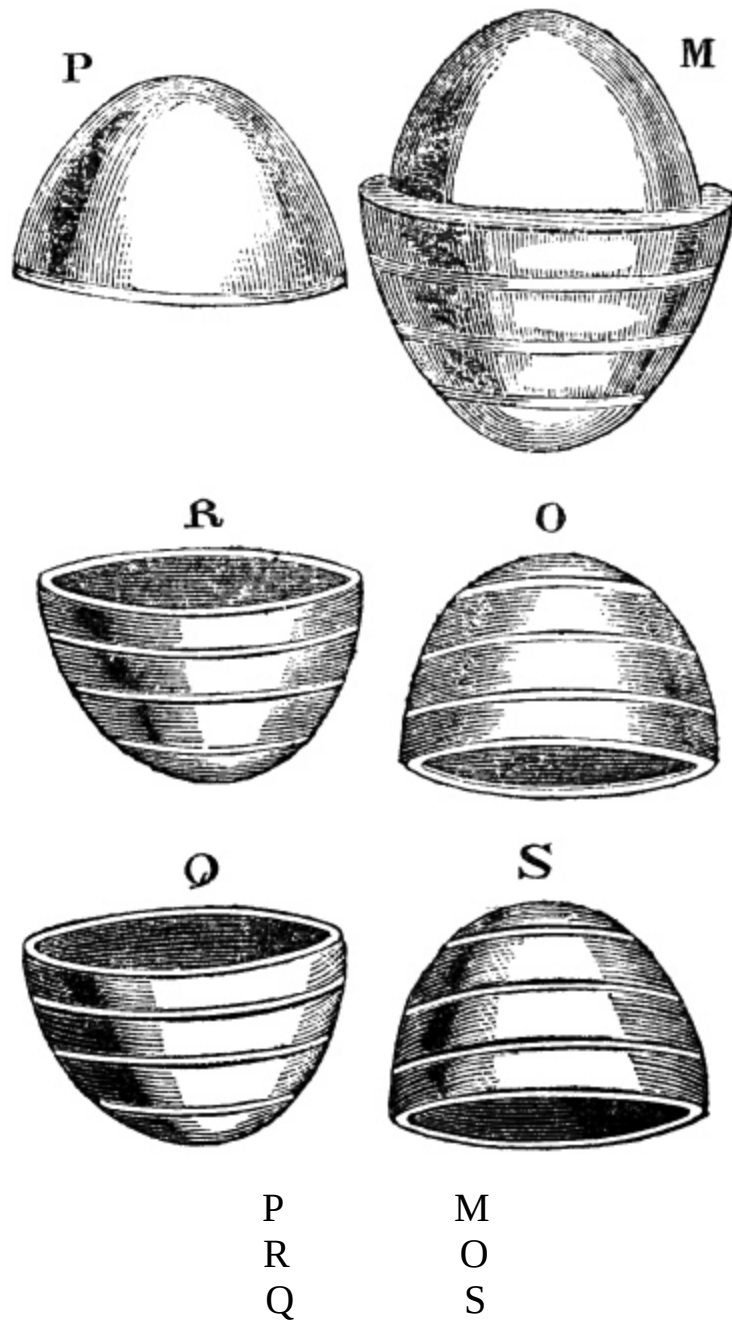
When you go to show this trick, desire any in the company to lend you a sixpence, saying, "*you will return it safe*"; requesting, withal, that none will meddle with any thing they see, unless you desire them, lest they prejudice you and themselves. Then take the cap off your box, and bid any one see it and feel it, that there be no mistrust; so likewise take the box entire, holding your fore-finger on the bottom, and your thumb on the upper part, turning it upside-down, and say,—"*You see here is nothing*;" then putting in the sixpence, put the cap over the box again; as the box stands covered upon the table, put your hand under the table, using some cant words; then take off the cap with your fore-finger and thumb, so as you pinch the innermost box with it and set it gently on the table; then put the dead quicksilver out of the lower part, into your hand, turning the box with the bottom upwards, and stirring it about with your fore-finger; then say, "*Here you see it melted, now I will put it in again, and turn it into single pence*;" suddenly take the cap as you took it off, and return it again; bid them blow on it; then take off the cap as you did before, only pinching the uppermost lid in it, and setting it upon the table; hold the box at the top and bottom, with your fore-finger and thumb; then put the six single pence, after they are viewed, and seen to be so, in again, and return the cap as before, saying, "*Blow on it if you would have it in the same form you gave it me*;" then taking the cap by the bottom, holding the box as before, put out the sixpence, and return the box into your pocket. This is a very good sleight, if well performed.

Trick upon the Globe-Box.

This is a trick not inferior to the best that is shown with boxes: it is a box made of four pieces, and a ball, so big as is imagined to be contained therein; the ball serves in the same way as the egg does in the egg-box, only to deceive the hand and eye of the spectators: this ball, made of wood or ivory, is thrown out of the box upon the table, for every one to see that it is substantial; then, putting the ball into the box, and letting the standers-by

blow on the box, taking off the upper shell with your fore-finger and thumb, there appears another, and of another colour,—as red, blue, yellow, or any variety of colours upon each ball that is so imagined to be, which, indeed, is no more than the shell of wood, ingeniously turned and fitted for the box, as you may see in the following figures:—





L is the outer shell of the globe, taken off the figures M N, an inner shell; O, the cover of the same; P, the other inner shell; Q, the cover of the same; R, the third shell; S, that which covers it. These globes may be made of more or less variety, according to the wish of the operator.

Trick with the Funnel.

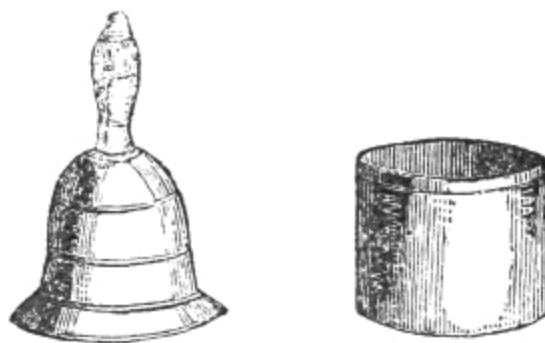


You must get a double funnel; that is, two funnels soldered one within the other, so that you may, at the least end, pour in a quantity of wine or water; this funnel you must have ready filled beforehand, with whatsoever liquor you please, and call for some of the same kind; then draw the funnel, and, setting your middle finger under the bottom of it, bid somebody (or do it yourself) pour it full, and drink it up before them, and turn the broad end of the funnel downwards, saying, "*Gentlemen, all is gone;*" and in a thrice turn yourself about, and, in turning, pronounce some term of art; withdraw your finger from the narrow end, and let the liquor out between the funnels, and it will be thought to be that which you drank out of the funnel, and so you may persuade them it is the same.

The Magical Bell and Bushel.

This bushel must be turned neatly, like the egg-boxes, so that they cannot find out where it opens; and you must have a false lid to clap on and off, and upon that false lid glue some bird-seed; and then you must have a true lid made to clap neatly upon the false one; now you must have your artificial bell to show with your bushel.

You may make your bell with wood or brass; but it must be made to unscrew at top, that it may hold as much seed as your bushel will when it is filled; and you must have the handle of your bell made with a spring, so as to let the seed fall down at your word of command.



The manner of using them is as follows:—Fill the top with bird-seed, before you begin to exhibit, saying, “*Ladies and gentlemen, you see I have nothing in my bell,*” (which they cannot do, if you hold it by the handle,) “*nor have I any thing in my bushel; therefore I will fill the bushel with bird-seed.*” In filling it, clap on the false lid, and no person will be able to discover the deception.

Now ask any one of the company to hold it in their hands whilst you command all the seed to appear under the magic bell. Then clap the true lid on, and ring the bell; accordingly, the seed will be gone out of the bushel into the bell, to the great wonder of the bystanders.

Out of an Empty Bag to bring upwards of an Hundred Eggs; and, afterwards, a living Fowl.

First, buy two or three yards of printed calico, or linen, and make a double bag, at the mouth of which, on that side next you, make four or five little purses, in each of which put two or three eggs, and do so till you have filled that side next to you. Have a hole made at one end of the bag, that no more than two or three eggs may come out at once. Then have another bag

like unto that exactly, that one must not be known from the other; put a hen into that bag, and hang it on a hook on the side you stand. The manner of performing it is thus: take the egg-bag and put both your hands in it; then turn it inside out, and say, "*Ladies and gentlemen, you see here is nothing in my bag;*" and in turning it again you must slip some of the eggs out of the purses, as many as you think fit: then turn your bag again, and show the company that it is empty, and in turning it command more eggs to come out. When all are come out but one, you are to take that egg and show it to the company, and then drop down the egg-bag and take up the hen-bag. Now, shake out the hen, pigeon, or other fowl. This, altogether, is a noble trick, if well executed.

Bonus Genius; or, Hiccius Doctius.



You must have the figure of a man made of wood, about the bigness of your little finger, the head whereof must be made to take off and put on at pleasure, by means of a wire that is in the neck; also, you must have a cloth cap, with a little bag within, to convey the head into; the bag must be neatly made, that it may not easily be perceived; show your man to the company, and say, "*Ladies and gentlemen, this I call my Bonus Genius;*" then show

the cap, saying, "*This is his coat;*" say moreover, "*look now as steadfastly as you can, nevertheless I will cozen you, for therefore am I come;*" then hold your cap above your face, and take your man in your right hand, and put his head through the hole of the cap, saying, "*Now he is ready to go on any message I have to send him,—to Spain or Italy, or where I will, but he must have somewhat to bear his charges;*" with that, pull out your right hand from under your cap, and therewith the body, but, privately putting your right hand into your pocket, as if you felt for money, where you may have the body, and take out your hand and say, "*There are three crowns for you, now begone;*" then turn the head, and say, "*But he will look about him before he goes;*" then say (setting your fore-fingers upon his crown), "*Just as I thrust my fore-finger down, so he shall vanish;*" and therewith, by the assistance of your left hand that is under the cap, convey his head into the little bag within the cap; then turn your cap about, and say, "*See here, he is gone;*" then take your cap and hold it up again, drawing the head out of the little bag, and say, "*Hic mecut genius;*" and in the meantime thrust the head through the hole of the cap, and, holding the head by the wire, turn it about presently, and put the head into your pocket.

To make a Knife leap out of a Pot.

Have a pot full of water standing on a table; then take a piece of whalebone, about three inches long,—let it be pretty stiff, that it may spring the better: take also a new stiff card, and fold it down the middle, longways; cut a hole through both folds at each end, half an inch, or more, from the ends; put one end of the whalebone in at one end of the card, bend it like a bow, and then put the other end of the whalebone into the other end of the card; set this in the pot, two inches deep in the water; then place the handle of the knife upon the uppermost part of the whalebone, with the point upwards, and use some words of art, as *Presto, begone!* when it springs out.

To turn a Box of Bird-seed into a living Bird.

You must have a box made on purpose, with a false lid. It must be turned neatly, like the egg-boxes, so that it cannot be perceived where it opens. You must likewise have a false lid to clap on and to take off. On that lid glue some bird-seed. Before you show the box to the company, put a bird into it; put on the false lid, and then show the box to the bystanders. It will seem to be full of seeds. Now put on the true lid, saying, "*Ladies and gentlemen, I command all the seeds out of my box, and order a living bird to appear;*" which it will do when the cover is taken off.



EXPERIMENTS WITH FIRE.

To produce a Carmine Red Flame.

The flame of spirit of wine may be coloured by the addition of various bodies which the spirit holds in solution, or which are mixed with it.

The flame of alcohol, or spirit of wine, is tinged red in the following manner: put into a small iron ladle, one part of *muriate of strontia*, and pour over it three or four of *alcohol*; then set it on fire with a candle, or a piece of burning paper; it will burn with a bright red carmine flame, especially if the mixture be heated, by holding the ladle over a candle or lamp, to cause the alcohol to boil rapidly.

The muriate of strontia left behind, after being again thoroughly dried, may be used again for the same purpose repeatedly.

An Orange-coloured Flame.

Put *muriate of lime* deprived of the water of crystallization, into an iron ladle, cover it with spirit of wine, and it will then burn in the manner stated.

To prepare the muriate of lime, dissolve common marble in muriatic acid, and evaporate the solution to perfect dryness.

To make Balloons with Soap and Water that catch Fire and detonate.

Fill with hydrogen gas a bladder with a cock fixed to it; to which adapt a copper tobacco pipe: dip the bowl, of the pipe into soap-lather; press

gently on the bladder filled with inflammable air, and a small soap bubble will issue, which, instead of falling to the ground, will rise most rapidly.

Instead of pure hydrogen gas only, fill the bladder with a mixture of one third of atmospheric air and two thirds of hydrogen gas: and, when they arise in the air, hold a candle to them, and the report will be like that of a pistol.

A Brilliant Blue Flame.

Mix boracic acid in spirits of wine, set the solution on fire, and the above effect will ensue.

An Emerald Green Flame.

Prepare nitrate of copper, thus:—let copper clippings or filings be dissolved in a sufficient quantity of nitric acid, of a moderate strength; when no further effervescence ensues, boil the acid gently upon the copper, until a pellicle appears; decant the solution, evaporate it slowly, and, when a very strong pellicle or skin appears, suffer it to crystallize. The salt is of a fine blue colour. Then cause the alcohol (spirit of wine) to burn upon this nitrate of copper, and it will exhibit a beautiful emerald green colour.

Loud Detonations, like the Discharge of Artillery.

Put into a bottle, not quite transparent, half an ounce of iron filings, over which pour four ounces of pure water; add one ounce of sulphuric acid, and shake the mixture.

If you apply a lighted paper tied to the end of a stick, near the top of the bottle, an instantaneous vivid flame will suddenly dart from the bottom, accompanied by a tolerably loud detonation.

Fifteen or twenty successive loud reports may likewise be obtained, by touching the aperture of the bottle with a lighted wax taper, fastened to a stick.

Though this experiment offers no very evident danger to the operator, it would be as well to envelop the bottle in a cloth, to guard against accidents.

A Well of Fire.

Add gradually one ounce, by measure, of sulphuric acid, to five or six ounces of water, contained in an earthenware basin, and add to it also, gradually, about three quarters of an ounce of granulated zinc. A rapid production of hydrogen gas will instantly take place. Then add, from time to time, a few pieces of phosphorus of the size of a pea. A multitude of gas-bubbles will be produced, which take fire on the surface of the effervescing liquid; the whole surface of the liquid will become luminous, and fire-balls, with jets of fire, will dart from the bottom, through the fluid, with great rapidity, and a hissing noise.

To make a Room seem all on Fire.

Take sal ammoniac half an ounce, camphor one ounce, aquæ vitæ two ounces; put them into an earthen pot in the fashion of a chamber-pot, but narrow something upon the top; then set fire to it, and the room will seem, to them that are in it, to be all on fire.

To walk on a hot Iron Bar, without Danger of Burning.

Take half an ounce of camphor, dissolve it in two ounces of aquæ vitæ, add to it one ounce of quicksilver, one ounce of liquid storax, which hinders the camphor from firing; take also two ounces of *hematitis*, a red stone, to be had at the druggist's; and when you buy it, let them beat it to powder in their great mortar, for it is so very hard that it cannot be done in a small one;

put this to the above-mentioned composition, and, when you intend to walk on the bar, you must anoint your feet well therewith, and you may walk over without danger. By this you may wash your hands in boiling lead.

To Eat Fire, and blow it up in your Mouth with a Pair of Bellows.

Anoint your tongue with liquid storax, and you may put a pair of tongs into your mouth, red hot, without hurting yourself, and lick them till they are cold. By the help of this ointment, and by preparing your mouth thus,



you may take wood-coals out of the fire, and chew them as you would bread; dip them into brimstone powder, and the fire will seem more strange. The sulphur puts out the coal, and shutting your mouth close puts out the sulphur. You may then chump the coals and swallow them, which you may do without offence to the body. In the same way, if you put a piece of lighted charcoal into your mouth, you may suffer a pair of bellows to be blowing in your mouth continually, and receive no hurt; but your mouth

must be quickly cleaned, otherwise it will cause a salivation. It is a very dangerous thing to be done; and although those that practise it use all the means they can to prevent danger, yet I never saw any of those fire-eaters that had a good complexion. Some put bole ammoniac into this recipe, a thing which spoils the whole composition, and so leave out hematitis and liquid storax; but let them beware how they use it.

To Light a Candle by a Glass of Water.

Take a little piece of phosphorus of the size of a pinhead, and with a piece of tallow stick it on the edge of a drinking-glass. Then take a lighted candle, and, having blown it out, apply it to the glass, when it will immediately be lighted. You may likewise write with a bit of phosphorus on paper some horrible words, which will appear awful when the candle is withdrawn from the room.

Fulminating Powder.

Take three parts of saltpetre, two parts of salt of tartar, and one part of sulphur pounded and mixed together; heat sixty grains of this composition in a spoon, and it will fly away with a fearful noise, like thunder. A grain by weight of this powder, put into a tobacco-pipe, will cause a loud report and much merriment, particularly if the smoker is not aware of the trick put upon him.

To set fire to a Combustible Body by the Reflection of two concave Mirrors.

The rays of a luminous body, placed in the focus of a concave mirror, being reflected in parallel lines, if a second mirror be placed diametrically opposite the first, it will, by collecting those rays in its focus, set fire to a combustible body.

Place two concave mirrors at about twelve or fifteen feet distance from each other, and let the axis of each of them lie in the same line. In the focus of one of them place a live coal, and in the focus of the other some gunpowder. With a pair of double bellows, which make a continual blast, keep constantly blowing the coal, and, notwithstanding the distance between them, the powder will presently take fire.

It is not necessary that these mirrors be of metal or glass; those made of wood or pasteboard, gilded, will produce the explosion, which has sometimes taken effect at the distance of fifty feet, when mirrors of eighteen inches or two feet diameter have been used.

To give the Faces of the Company the Appearance of Death.

Dissolve some common salt in spirits of wine, and pour the whole on some saffron in a saucer. Take away the candle from the room, and set fire to the spirit with a little burning tow. The countenances of every person in the room will, by this light, put on a cadaverous, deadly appearance, the fairest complexion appearing green, whilst the red of the lips and cheeks will be of an olive hue.

To dispose two little Figures, so that one shall light a Candle, and the other put it out.

Take two little figures of wood or clay, or any other materials you please, only taking care there is a little hole at the mouth of each. Put in the mouth of one a few grains of bruised gunpowder, and a little bit of phosphorus in the other, taking care that these preparations are made beforehand; then take a lighted wax candle, and present it to the mouth of the figure with the gunpowder, which, taking fire, will put the candle out; then present your candle, having the snuff quite hot, to the other figure; it will light again immediately by means of the phosphorus. You may propose the same effect to be produced by two figures, drawn on a wall with a pencil or a coal, by applying, with starch or a wafer, a few grains of bruised

gunpowder to the mouth of one, and a bit of phosphorus to the mouth of the other.

To construct a Lantern which will enable a Person to read by Night, at a great Distance.

Make a lantern of a cylindric form, or shaped like a small cask placed lengthways, so that its axis may be horizontal, and fix in one end of it a parabolic or spheric mirror, so that its focus may fall about the middle of the axis of the cylinder. If a small lamp or taper be placed in this focus, the light, passing through the other end, will be reflected to a great distance, and will be so bright, that very small letters on a remote object may be read, by looking at them with a good telescope. Those who see this light, if they be in the direction of the axis of the lantern, will think they see a large fire.



TRICKS WITH STRINGS, KNOTS, &c.

To cut a Lace asunder in the Middle, and to make it whole again.

Provide a piece of the lace which you mean to cut, or at least a pattern like the same, one inch and a half long, and keep it double privately in your left hand, between some of your fingers, near to the tips thereof; take the other lace which you mean to cut, still hanging about your neck, and draw down your said left hand to the bout thereof, and, putting your own piece a little before the other, the end, or rather middle, whereof you must hide betwixt your fore-finger and thumb, make the eye or bout which shall be seen of your own pattern; let a bystander cut the same asunder, and it will be surely thought that the other lace is cut; which, with words and fretting, you shall seem to renew, and make whole again. This, if it be well handled, will seem very wonderful.

To burn a Thread and make it Whole again with the Ashes.

Take two pieces of thread of one foot in length each: roll one of them round like a small pea, which put between your left fore-finger and thumb. Now, hold the other out at length, between the fore-finger and thumb of each hand, holding all your fore-fingers daintily; then let one cut the same asunder in the middle; when that is done, put the tops of your two thumbs together, and so shall you with less suspicion receive the thread which you hold in your right hand into your left, without opening your left finger and thumb. Then holding these two pieces as you did before it was cut, let these two be cut also asunder in the middle, and they conveyed again as before, until they be very short; then roll all these ends together, and keep that ball of thread before the other in the left hand, and, with a knife, thrust the same into a candle, where you may hold it until the said ball of thread be burnt to

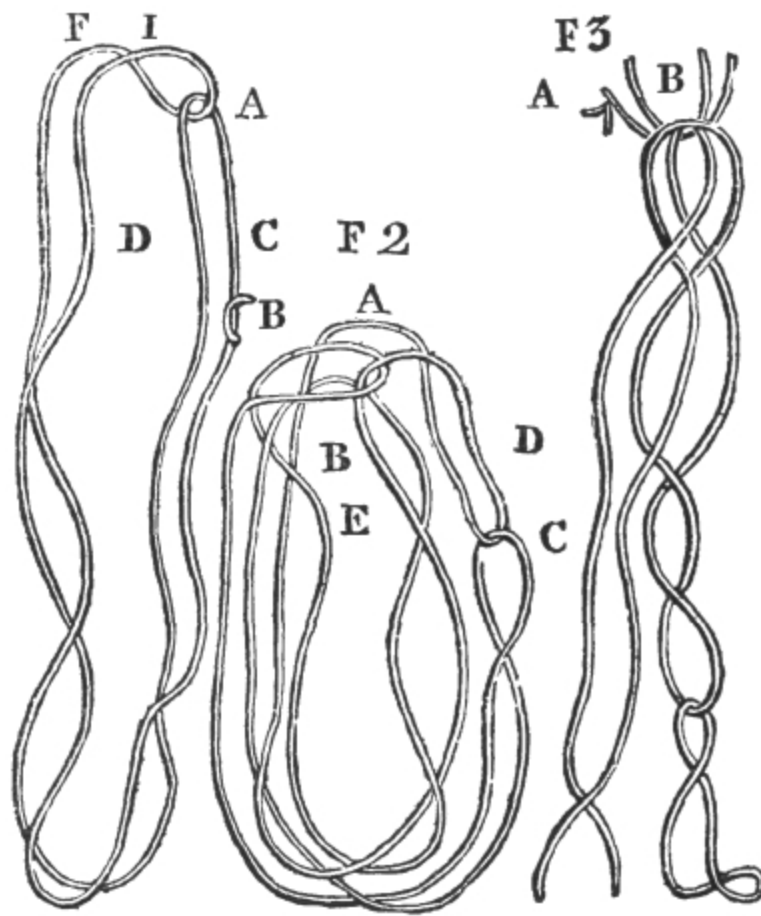
ashes; then pull back the knife with your right hand, and leave the ashes, with the other ball betwixt your fore-finger and thumb of your left hand, and, with the two thumbs and two fore-fingers together, take pains to rub the ashes till your thread be renewed, and draw out that thread at length, which you lay all this while betwixt your fore-finger and thumb. This is not inferior to any juggler's trick, if it be well handled; for if you are so perfect in legerdemain as to bestow the same ball of thread, and to change it from place to place, betwixt your other fingers, as may be easily done, then it will seem very strange.

To pull many Yards of Ribbon out of the Mouth.

Jugglers get money from maids, by selling laces by the yard, putting into their mouths one round bottom, as fast as they pull out another, and, at the exact end of every yard, they tie a knot, so as the same rests upon their teeth; then they cut off the same, and so the beholders are doubly and trebly deceived, seeing as much lace as will fill a hat, and the same of what colour they like, to be drawn out of the mouth; and yet the juggler talks as though there was nothing in his mouth.

To cut a Piece of Tape into four Parts, and make it Whole again with Words.

Take a piece of narrow white tape, about two or three yards long. First present it to view to any that may desire it; then tie both the ends of it together, and take one side of it into one hand, so that the knot may be about the middle of one side, and, using some circumstantial words to beguile the spectators, turn one hand about toward yourself, and the other from you; so shall you twist the tape once, then clap the ends together, and then, if you slip your finger and thumb of each hand between the tape, almost as one would hold a skein of thread to be wound, this will make one fold or twist, as thus appears, where A signifieth the twist or fold—B the knot. Then



in like manner make a second fold, about the line D C, as you may see by the second figure, where B signifieth the knot, C the first fold, and A the second fold; then hold the fore-finger and thumb of your left hand upon the second twist, and upon the knot also, and the fore-finger and thumb of your right hand upon the first fold C, and desire some one of your spectators to cut all asunder with a sharp knife at the cross line E D; when it is cut, hold still your left hand, and let all the ends fall that you hold in your right hand, for there will be a show of eight ends, four above and four below, and so the strings will be thought to be cut into four parts, as may be seen by the third figure; then gather up the ends that you let fall into your left hand, and deliver two of the ends (seeming to take them at random) unto two several persons, bidding them hold them fast, still keeping your left-hand fingers upon the twists or folds; then with both your hands seem to tumble all the ends together that you had in your left hand, twist out the slips and pieces,

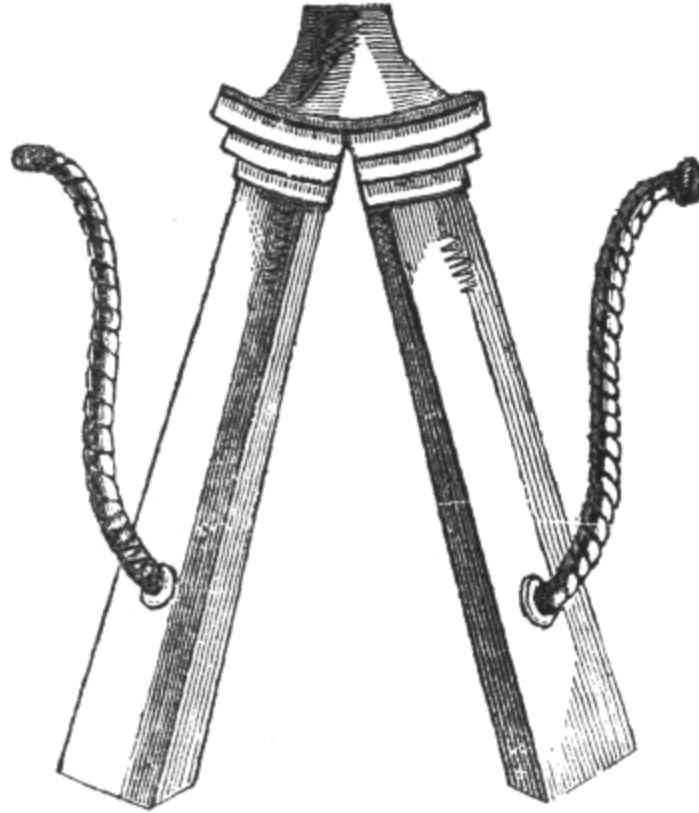
which are three, as you may see at A and B in the third figure,—twist them all, I say, into a little ball, and conceal it between some of your fingers of your left hand, and crumble thereon another confused heap, and, after some words said, with your right hand deliver this confused heap unto any of the company, bidding them hold it fast, saying, “*Hulla passa;*” then bid them look on it, and while they are greedily looking after the event, you may with ease convey the ball or roll of ends into your pocket; so it will be thought that you have made it whole by virtue of your words. This is an excellent trick, if it be gracefully handled: it cost me a great deal of trouble and time to find it out.

To unloose a Knot upon a Handkerchief by Words.

Make a plain loose knot with two ends of a handkerchief, seeming to draw the same very hard. Hold fast the body of the handkerchief near to the knot, with your right hand, pulling the contrary end with your left. Then close up handsomely the knot, which will be somewhat loose, and pull the handkerchief so with your right hand, that the left hand may be near the knot; then will it seem to be a true and firm knot; and to make it appear more assuredly to be so indeed, let a stranger pull at the end which you hold in your left hand, while you hold fast the other in your right hand, and then holding the knot with your fore-finger and thumb, and the lower part of your handkerchief with your other finger, as you hold a bridle when you would with one hand slip up the knot and lengthen the reins. This done, turn your handkerchief over the knot with your left hand; in doing whereof you must suddenly slip out the end or corner, putting up the knot of your handkerchief with your fore-finger and thumb, as you would put the aforesaid knot of your bridle. Then deliver the same, covered and wrapped within the midst of the handkerchief, to one to hold fast, and, after the pronounciation of some words of art, take the handkerchief and shake it, and it will be loose.

To draw a Cord through the Nose.

This trick is called the bridle, being made of two alder sticks—



Through the hollowness thereof is placed a cord, the same being put upon the nose, like a pair of pincers or tongs; the cord which goeth round the same being drawn to and fro, the beholders will think the cord goes through your nose very dangerously; the knots at the end of the cord, which stay the same from being drawn out of the stick, must not be put at the very top, for that must be stopped up, but half an inch beneath each end; and so, when it is pulled, it will seem to pass through the nose. Then you may take a knife and seem to cut the cord asunder, and pull the bridle from your nose.

To take three Button-Moulds off a String.

Take two little whipcords, of two feet long each, double them equally, so as there may appear four ends; then take three button-moulds (the hole of one of them must be bigger than the rest), and put one button-mould upon the eye or bout of one cord, and another on the other cord; then take the button-mould with the greatest hole, and let both the bouts be hidden therein, which may be the better done if you put the eye, or bout, of the one



into the eye or bout of the other; then put the middle button upon the same, being doubled over his fellow: so will the heads seem to be put over the two cords. You may loose them as you like, and make it seem manifest to the beholders, who may not see how they are done; but that the buttons are put upon the two cords, without any fraud. Then must you seem to add a more effectual binding of those buttons to the strings, and make one half of a knot with one of the ends of each side, which is for no other purpose, but that, when the buttons are taken away, the cords may be seen in the case which the beholders suppose to be in before; for when you have made your half knots, which in any wise you must not double to make a perfect knot, you must deliver into the hands of some bystanders these two cords—namely, two cords evenly set to one hand, and two in the other, and then with a wager begin to pull off the buttons, which, if you handle nimbly, and in the end cause him to pull his two ends, the two cords will show to be placed plainly, and the buttons to have come through the cords.

OPTICAL ILLUSIONS.

The Multiplying Mirror.

A mirror is to be made as follows: first make a hoop or fillet of wood about the width of a half-crown piece, and let the thickness be about a quarter of an inch. In the middle of this hoop fasten a bottom of wood or brass, and place several hooks of the size of small peas. Then open the one side of this bottom, set in a piece of crystal glass, and fasten it in the hoop close to the bottom;



then take a quantity of quicksilver, and put so much into the hoop as will cover the bottom; then let into the hoop another piece of crystal glass fitted thereto, and cement the sides that the quicksilver may not run out, and it is done,—the use whereof I shall not insist upon, since he that is versed in optical illusions will better conceive it, than any words of mine can either direct or assist him.

The Magic-Lantern.

This lantern is called magical on account of the formidable apparitions that, by virtue of light, show upon the white wall of a dark room. The body of it is generally made of tin, and of the shape of a lamp. Towards the back is a concave looking-glass of metal, which may either be spherical or parabolical, and which, by a groove made in the bottom of the lantern, may be either advanced nearer or put farther back from the lamp, in which is oil or spirit of wine; and the match ought to be a little thick, that when it is lighted it may cast a good light that may easily reflect from the glass to the fore part of the lantern, where there is an aperture with a prospective in it, composed of two glasses, that make the rays converge and magnify the object.

When you mean to make use of this admirable machine, light the lamp, the light of which will be much augmented by the looking-glass at a reasonable distance. Between the fore part of the lantern and the prospective glass you have a trough, made on purpose, in which you are to run along a flat frame, with transparent colours on glass. Then all these little figures, passing successively before the prospective glass, through which passes the light of the lamp, will be painted and represented with the same colours upon the wall of a dark room, and in a gigantic and monstrous manner.

By this lantern you may show birds, beasts, and all sorts of fish, &c. &c. &c.

The Phantascope.

The Phantascope is a magic-lantern improved; that is, it will magnify or diminish the figures it is to reflect, without impairing their neatness.

This instrument is composed of a tin box, surmounted by a kind of chimney, pierced on one of its faces with a hole of a diameter corresponding with that of the cylinders, sliding into one another like the tubes of an opera-glass: this tube is adjusted by means of a slide established above the aperture made in the phantascope. The part of the adjoining cylinder is so

adjusted to receive alternately wooden tablets, in which are opened cavities to contain the figures painted upon glass. The flat side of the half ball is placed in that tube, nearly close enough to touch the painted figures: this half ball is called the object glass.

In the second cylinder is found a magnifying-glass, convex on both sides, the focus whereof is double to that of the half ball, and one third of its diameter: this glass is called “the eye-glass.” It is by means of drawing those glasses close that the objects are magnified; and by removing them at a distance, that the figures are made to look smaller. Inside of the box is placed one of those lamps called “argand,” of a wide diameter, with a concave mirror, to throw as much light as possible upon the transparent figures.

To exhibit the phantasmagoria, select an apartment opening into a second room, and the door of which is at your disposal: take a wooden frame, exactly of the same dimensions as that door, upon which fasten, with brass nails, a sheet of fine linen, in preference to calico. A moment before you place the frame, wet it equally all over, to cause the thread of the cloth to swell, and to fill all interstices.

When you have proved the above-mentioned instrument, prior to your using it, make sure that the lamp does not smoke, but that it gives a vivid light; that the concave mirror is very bright; that the magnifying-glass is dry and clean. (You dry these with a piece of soft leather and some whitening, so fine as not to scratch. The painted glasses are cleaned with a piece of very fine cloth, made quite hot, or of cambric, taking care not to rub off the painting.) You then place the wooden ruler with the reversed figure in the slide, opened purposely to receive it, in the first cylinder; then, the figures being reflected on a whitewashed wall, you regulate the distance of the magnifying-glasses, in order to have the paintings seen in all their perfection at a regular fixed distance. Your instrument is then ready for the exhibition.

To this effect, you present yourself facing the framed linen, with the phantoscope suspended before you, and held by two braces across your shoulders. In order that the figures may decrease in the eyes of the spectators of the other side, you must advance slowly near the cloth,

drawing the two magnifying-glasses at a distance from each other, but very gently, that the figures should remain steady.

If, on the contrary, you should wish them to appear magnified, gradually draw back, with the same precaution, and bring the glasses nearer each other; when they are quite close, the figures seem to be coming towards the spectators, and to proceed forward.

When you change the paintings, you must hide the eye-glass behind an opaque and black body. The knowledge of the management of the glasses, and the accuracy of the proceeding backward and forward, upon which depends entirely the good effect of the exhibition, may be acquired by a few days' practice previous to the exhibition being made in company.

Notwithstanding the box for the phantascope, if made of wood, will be heavier than a tin one, yet it is to be preferred, on account of its communicating much less heat than the tin.

Such persons as wish to paint their figures should use a varnish that does not spread on the glass; thick copal varnish will answer the purpose.

The Enchanted Mirrors.

Make a box of wood, of a cubical figure, of about fifteen inches every way. Let it be fixed on a pedestal, at the usual height of a man's head. In each side of this box, let there be an opening of an oval form, ten inches high, and seven wide.

In this box place two mirrors, with their backs against each other; let them cross the box in a diagonal line, and in a vertical position. Decorate the openings in the sides of this box with four oval frames, and transparent glasses, and cover each of them with a curtain, so contrived that they may all draw up together.

Place four persons in front of the four sides, and at equal distances from the box, and then draw up the curtains, that they may see themselves in the mirrors; when each of them, instead of his own figure, will see that of the person who is next him, and who, at the same time, will seem to him to be placed on the opposite side. Their confusion will be the greater, as it will be

very difficult for them to discover the mirror contained in the box. The reason of this phenomenon is evident; for, though the rays of light may be turned aside by a mirror, yet they always appear to proceed in right lines.

The Wonderful Phantoms.

In a partition make two apertures of a foot high, and ten inches wide, and about a foot distant from each other. Let them be at the common height of a man's head; and in each of them place a transparent glass, surrounded with a frame, like a common mirror.

Behind this partition place two mirrors, inclined to it in an angle of forty-five degrees (that is, be halfway between a line drawn perpendicular to the ground and its surface); let them be both eighteen inches square; let all the space between them be enclosed by boards, or pasteboard, painted black, and well closed, that no light may enter. Let there be also two curtains to cover them, which may be drawn aside at pleasure.

When a person looks into one of these supposed mirrors, instead of seeing his own face, he will perceive the object that is in front of the other; so that, if two persons present themselves at the same time before these mirrors, instead of each one seeing himself, they will reciprocally see each other.

Note.—There should be a sconce, with a candle placed on each side of the two glasses in the wainscot, to enlighten the faces of the persons who look in them; otherwise this experiment will have no remarkable effect.

This curious trick may be considerably improved, by placing the two glasses in the partition in adjoining rooms; and, a number of persons being previously placed in one room, when a stranger enters the other, you may tell him his face is dirty, and desire him to look in the glass, which he will naturally do; and on seeing a strange face he will draw back; but, returning to it, and seeing another, and another, like the phantom kings in Macbeth, what his surprise will be is more easy to conceive than express. After this, a real mirror may be privately let down on the back of the glass, and, if he can be prevailed to look in once more, he will then, to his further

astonishment, see his own face, and may be told, perhaps persuaded, that all he thought he saw before was mere imagination.

How many tricks, less artful than this, have passed in former times for sorcery, and pass at this time, in some countries, for apparitions!

Note.—When a man looks in a mirror that is placed perpendicular to another, his face will appear entirely deformed. If the mirror be a little inclined, so as to make an angle of eighty degrees (that is, one ninth part from the perpendicular), he will then see all the parts of his face, except the nose and forehead. If it be inclined to sixty degrees (that is, one third part), he will appear with three noses and six eyes: in short, the apparent deformity will vary at each degree of inclination; and when the glass comes to forty-five degrees (that is, half way down), the face will vanish. If, instead of placing the two mirrors in this situation, they are so disposed that their junction may be vertical, their different inclinations will produce other effects, as the situation of the object relative to these mirrors is quite different.

The Real Apparition.

Behind a partition place, in a position somewhat oblique, a concave mirror, which must be at least ten inches in diameter, and its distance from the partition equal to three fourths of the distance of its centre.

In the partition, make an opening of seven or eight inches, either square or circular: it must face the mirror, and be of the same height with it. Behind this partition place a strong light, so disposed, that it may not be seen at the opening, and may illumine an object placed near it, without throwing any light on the mirror.

Beneath the aperture in the partition, place an object that you intend shall appear on the outside of the partition, in an inverted position, and which we will suppose to be a flower. Before the partition, and beneath the aperture, place a little flower-pot, the top of which should be even with the bottom of the aperture, that the eye may see the flower in the same position as if its stalks came out of the pot.

Take care to paint the space between the back part of the partition and the mirror black, to prevent any reflections of light from being thrown on the mirror; in a word, so dispose the whole that it may be as little enlightened as possible.

When a person is placed at the point of vision, he will perceive the flower that is behind the partition, at the top of the pot; but, on putting out his hand to pluck it, he will find that he attempts to grasp a shadow.

Observations.

The phenomena that may be produced by means of concave mirrors are highly curious and astonishing. By their aid spectres of various kinds may be exhibited. Suppose, for example, you were to tell any one, that at such an hour, and in such a place, he should see the apparition of an absent or deceased friend (of whose portrait you are in possession); in order to produce this phantom, instead of the hole in the partition, there must be a door which opens into an apartment to which there is a considerable descent. Under that door you are to place the portrait, which must be inverted and strongly illuminated, that it may be properly reflected by the mirror, which must be large and well polished. Then, having introduced the spectator at another door, and placed him in the proper point of view, you suddenly throw open the first door, when, to his great astonishment, he will immediately see the apparition of his friend.

It will be objected, perhaps, that this is not a perfect apparition, because it is only visible at one point of view, and by one person. But it should be remembered, that it was an established maxim in the last century, that a spectre might be visible to one person, and not to others. So Shakespeare makes both Hamlet and Macbeth see apparitions that were not visible to others, present at the same time. It is not unlikely, moreover, that this maxim took its rise from certain apparitions of the kind, *that were raised by the monks, to serve some purposes they called religious*; as they alone were in possession of what little learning there then was in the world.

***To draw a Deformed Figure, which will appear well proportioned
from a certain Point of View.***

Draw any thing you may fancy on a thin white pasteboard; then prick it; afterwards put the same on an horizontal surface, which we will suppose to be another pasteboard; put a lighted candle behind that drawing, and draw on the horizontal surface the lines given by the light; this will give a deformed design. This being done, take away the drawing that was pricked, and the candle; then place your eye where the light was, and you will see your drawing assume a regular form.



CHEMICAL CHANGES.

To change the Colour of a Rose.

Burn some sulphur under the rose, by holding underneath it a lighted bundle of matches. This will fade the red colour, and make the flower turn white or blanched. Its primitive colour will return in about a couple of hours.

To turn Water into Wine.

Take four glasses; rub one in the inside with a piece of moist alum: let the second have a drop of vinegar in it; the third empty. Let the fourth be filled with clean water; in your mouth put a clean rag, with ground basil tied close in it, the bulk not to be bigger than a small nut, which must lie between your hind teeth and your cheek; then take some of the water that is in the glass into your mouth, and return it into the glass that hath the drop of vinegar in it, which will cause it to turn the perfect colour of sack wine; then turn it in your mouth again, and chew the rag of basil between your teeth, and squirt the liquor into the glass, and it will have the perfect colour of claret; returning the basil into its former place, take the liquor into your mouth again, and presently squirt it into the glass you rubbed with alum, and it will have the perfect colour of mulberry wine.

Arbor Dianæ; or, the Silver Tree.

This name has been given to a beautiful and curious precipitation of silver, by means of mercury in an arborescent form. The experiment is

made by putting a soft amalgam of silver into six parts of a solution of nitrate of silver, and four of a solution of nitrate of mercury, or—

Amalgamate in a glass mortar one quarter of an ounce of pure mercury and half an ounce of fine silver. Insert in this amalgamation four ounces of pure nitric acid for the solution, and increase the whole by a pint of distilled water; mix it well, and keep it in a glass decanter well stopped. By introducing into this liquid a lump of soft amalgam of silver, the formation of the tree speedily takes place.

Or, if you wish to make the experiment on a smaller scale, pour about an ounce of the above liquid into a wine-glass, into which insert a piece of soft amalgam of silver, the size of a pea, and the effect required will immediately be produced. The silver, separating in the form of prismatic needles, will arrange itself so as to assume an arborescent appearance.

The Lead Tree.

The precipitation of a solution of acetate of lead by means of a piece of zinc, so as to form what is commonly called the lead tree, is perfectly analogous to this theory, but is too well known to require illustration.

The Tree of Mars.

Dissolve iron filings in aqua fortis moderately concentrated, till the acid is saturated; then pour gradually into the solution a solution of fixed alkali, commonly called oil of tartar. A strong effervescence will take place, and the iron, instead of falling to the bottom of the vessel, will afterwards rise, so as to cover its sides, forming a multitude of ramifications heaped one upon the other, which will sometimes pass over the edge of the vessel, and extend themselves on the outside, with all the appearance of a plant. If any of the liquor is spilt, it must be carefully collected, and be again put into the vessel, where it will form new ramifications, which will contribute to increase the mass of the vegetation.

To form a Metallic Tree, in the Shape of a Fir.

In a small globular bottle with a tall neck, dissolve one ounce of fine silver in three ounces of nitric acid; put the bottle into a sand bath, and let it remain over a slow fire till the liquid be half reduced; then add three ounces of good white-wine vinegar, a little warmed; shake the mixture, and place the bottle in any place where it is not likely to be disturbed: in about a month's time a beautiful metallic fir will appear, ramifying to the very surface of the liquid.

To make a Gold or Silver Tree, to serve as a Chimney Ornament.

Make an amalgam of a small portion of fine gold, and ten times the quantity of purified cinnabar.

Grind and wash the amalgam in water, till no black colour appears: when perfectly clean, dry the amalgam, and insert it into a glass retort; place it in a sand bath over a slow fire, which must be preserved ignited for a day or two; gradually increase the heat, to expel the mercury from the gold, whose metallic vegetation will be perfectly in proportion to the mercury which escapes. When the fire is extinguished you will find the mercury expelled into the receiver, and the gold remaining in the retort will be soft and malleable, possessing the purest appearance; the mass will have sprouted branches, in perfect imitation of small shrubs and trees of different forms and height, which may be separated from the heap, serving as their base, when taken from the retort. These will allow even of being made red hot without injury to their appearance, or fear of decomposition, and may be used as ornaments for any length of time.

For the silver, use the same proportion of metal as for the gold tree.

Sympathetic or Secret Inks.

Sympathetic or secret inks are those fluids which, when written with on paper, are invisible when dry, but acquire colour by heating the paper, or by

applying to the invisible writing another chymical agent. These phenomena arrested particularly the attention of the old chymists, and accordingly, in their fanciful way, they called them *sympathetic inks*.

The writing made with this ink may, therefore, at pleasure be made to become visible or invisible successively, by alternately warming and cooling, if care be taken not to expose the paper to a greater degree of heat than is necessary to make the invisible writing legible.

Preparation of Green Sympathetic Ink.

Put into a matrass one part of cobalt or zaffree, and four ounces of nitro-muriatic acid. Digest the mixture with a gentle heat until the acid dissolves no more; then add muriatic of soda, equal in quantity to the cobalt employed, and four times as much water as acid, and filter the liquor through paper.

Blue Sympathetic Ink.

This ink, which may be used like the preceding, is prepared in the following manner. Take one ounce of cobalt reduced to powder, put it into a Florence flask, and pour over it two ounces of pure nitric acid. Expose the mixture to a gentle heat, and, when the cobalt is dissolved, add, by small quantities, a solution of potash, until no more precipitate ensues. Let these precipitates subside: decant the super-natant fluid, and wash the residuum repeatedly in distilled water, until it passes tasteless; then dissolve it in a sufficient quantity of distilled vinegar, by the assistance of a gentle heat, taking care to have a saturated solution, which will be known by part of the precipitate remaining undissolved after the vinegar has been digested on it for some time.

Yellow Sympathetic Ink.

Neutralize muriatic acid with brown oxide of copper: the solution is of a dark olive-green colour, and by evaporation yields crystals of a grass-green colour of muriate of copper, which, when dissolved in ten parts of water, forms this ink, and may be employed as before stated.

Purple Sympathetic Ink.

Dissolve cobalt or zaffre in nitric acid, add salt of tartar gradually and in small quantities to avoid too powerful an effervescence, let the precipitate subside, and, having drawn off the super-natant clear liquid, add a sufficient quantity of water, when the ink will be fit for use.

Rose-coloured Sympathetic Ink.

Dissolve zaffre in nitric acid, to which add saltpetre well purified, and you will possess a rose-coloured ink, having the same properties as the preceding.

Application of the Secret Inks.

Write any unimportant matter with common ink, and let the lines be tolerably wide apart; then between these lines write the communication you wish to make with any of the above invisible inks. Your correspondent, by holding the paper before the fire, will be speedily enabled to peruse the letter; the characters will again become invisible when the paper has cooled. The writing in common ink will serve to lull the suspicions of those who might intercept the letter.

With this kind of ink, some very ingenious and amusing tricks may be performed.

A Drawing which alternately represents Winter and Summer Scenes.

Draw a landscape, and delineate the ground, the trunks, and branches of the trees, with the usual water-colours employed for that purpose, tracing the grass and trees with sympathetic ink. By these means you will have a drawing, which, at the common temperature of the atmosphere, will represent a winter piece; but if it be exposed to a proper degree of heat, not too strong, you will see the ground covered with verdure, and the trees suddenly wrapped with a beautiful foliage; replace the picture in the cool, and the dreary aspect of winter re-appears. The colours must be lighter for the back grounds.

Screens painted in this manner were formerly made at Paris. Those to whom they were presented, if unacquainted with the artifice, were astonished to find, when they made use of them, that the views they exhibited were totally changed. Persons not understanding drawing may amuse themselves in this manner, by colouring an outlined engraving representing winter, following exactly the process enjoined for the above.

Demonstration of the various Strata of Earth which cover the Globe.

In a middling-sized glass bottle, three parts full of clear water, infuse pure alumine precipitated by ammonia, till the bottle is nearly filled.—Expose the bottle to the cold air in frosty weather; or, at other times, to artificial frost. At the moment when the frigorific effect ensues, the alumine will divide all over the surface of the water, and form itself into separate and very regular strata.

To freeze Water in the Midst of Summer, without the Application of Ice.

Take eleven drachms of muriate of ammonia, ten of nitrate of potash, and sixteen of sulphate of soda; reduce each of these salts separately to a fine powder, and mix them gradually in a glass, or rather in a thin metal vessel, with five ounces of water (the capacity of the vessel should be only just large enough to hold the materials): the result will be, that, as the salts dissolve, cold will be produced, and a thermometer immersed in the mixture will sink at or below freezing. A little water, about half an ounce, in a test tube, when immersed in the mixture during its solution, becomes frozen in about ten minutes.

The salts employed in this experiment may be recovered from the solution, if the sulphate of soda be omitted, by evaporating the water, to serve again any number of times. Five parts of muriate of ammonia, five of nitre, and eight of sulphate of soda, mixed with sixteen of water, at the usual temperature, sink the thermometer from 50 to about 10 degrees.

The salts, to produce their fullest frigorific effect, should be recently crystallized in fine powder, and contain as much water of crystallization as possible, but they should not be damp.

If to three parts of ice thus procured, you add four parts of potash, the thermometer will sink from 32° to 61° below Zero, or the freezing point; giving 93 degrees of cold. Snow or common ice, mixed with potash in the same proportions, would serve equally well; but the above experiments are given under the supposition that they are difficult to be procured.

MISCELLANEOUS TRICKS AND EXPERIMENTS.

To swallow a long Pudding made of Tin.

This pudding must be made of tin, consisting of twelve or thirteen little hoops, round, and in little ringlets, so that they may seem to fall one through another. It must have little holes made at the biggest end, that it may not hurt your mouth: hold this pudding, for so it is called, privately in your left hand, with the hole end uppermost, and with your right hand take a ball out of your pocket, and say, *“If there is any old woman that is out of conceit with herself, because her neighbours deem her not so young as she would be thought, let her come to me, for this ball is a present remedy;”* then seem to put the ball into your left hand, but let it slip into your lap, and clap the pudding into your mouth, which will be thought to be the ball that you showed them; then decline your head, and open your mouth, and the pudding will slip down at its full length, which, with your right hand, you may strike into your mouth again; doing this three or four times: then you may discharge it into your hand, and clap it into your pocket without any suspicion, by making three or four wry faces after it, as though it had stuck in your throat; and if you practise smiting easily on your throat, with your fist on each side, the pudding will seem to chink as if it was laying there; then say, *“Thus they eat puddings in High Germany; they fling them down their throats before their teeth can take possession of them.”*

An Artificial Spider.

Take a bit of burnt cork, as big as a pea, and give it the shape of a spider; make its legs with threads of hemp; put a grain of lead into the cork to give it some weight; then hang this artificial spider by a bit of gray

sewing silk (that is not twisted), between two bodies, the one electrified, and the other not; or between two bodies endowed with different electricities: it will go and come between these two bodies, and the movement of the legs will be seen as plainly as if it were a living spider.

To pass a Ring through your Cheek.

You must have two rings made of silver or brass, or what you please, of one size, colour, and likeness, saving that one must have a notch through it, and the other must be whole, without a notch; show the whole ring, but conceal that which hath the notch, and say, “*Now I’ll put this ring through my cheek,*” and privately slip the notch over one side of your mouth; then take a small stick, which you must have in readiness, and slip the whole ring upon it, holding your hand over it about the middle of the stick; then bid somebody hold fast the stick at both ends, and say, “*See this ring in my cheek,—it turns round;*” then, while you perceive them fasten their eyes upon that ring, on a sudden, whip it out, and smite upon the stick therewith; instantly concealing and whirling the other ring you hold your hand over, round about the stick. It will be thought that you have brought that ring upon the stick which was upon your cheek.

To cut a Hole in a Cloak, Scarf, or Handkerchief, and by Words to make it whole again.

To do this, you must have a piece of the same ready in your hand, the sample of that you intend to cut; then, amongst other tricks, clap your hand upon the place you intend to cut. Now drawing the false piece through, cause it to be cut off, and, griping your hand, show the hole from whence the piece came away which is in your hand. This is done by pretending to feel in your pocket for a needle and thread to sew it up again. But, drawing your hand out of your pocket, say “*I have no needle, but I have a charm that will do as well;*” so, muttering some words, bid them blow upon it, and, pulling your hand from the place, show it entire.

The Dancing Egg.

Three eggs are brought out; two of them are put on a table, and the third in a hat; a little cane is borrowed from one of the company, and it is shown about, to convince the spectators there is no preparation. It is then placed across the hat; the hat falls to the ground, and the egg sticks to it as if glued; the orchestra then plays a piece of music, and the egg, as if it was sensible of the harmony, twists about the cane from one end to the other, and continues its motions till the music stops.

Explanation.—The egg is fastened to a thread by a pin, which is put in lengthways; and the hole, which has been made to introduce this pin, is stopped with white wax. The other end of the thread is fastened to the breast of the person who performs this trick, with a pin bent like a hook; the cane, passing under the thread, near to the egg, serves for it to rest on. When the music begins, the performer pushes the cane from left to right, or from right to left; it then appears as if the egg ran along the cane, which it does not, being fastened to its thread; its centre of gravity remains always at the same distance from the hook that holds it; it is the cane which, sliding along, presents its different points to the surface of the egg.

N. B. To produce the illusion, and persuade the company that it is the egg which carries itself to the different points of the cane, the performer turns a little on his heel; by this means the egg receives a motion, which surprises the spectators, it remaining always at the same distance from the point to which it was fastened.

To make Three Figures dance in a Glass.

Take little hollow figures of glass about an inch and a half high, representing little children, or grown-up persons. These may be had of the glass-blowers; and, as they contain a body of air, are lighter than water. Immerse them in water contained in a glass of the following shape:—



This glass is about a foot or fifteen inches high, and covered with a bladder, which is tied fast over the top; a small quantity of air is to be left between the bladder and surface of the water; so, when you command them to walk down, press your hand hard upon the top, and they will immediately go down. Thus you may make them dance in the middle of the glass, at your pleasure; and, when you would have them go up to the top, take your hand away, when they will go up.

To shoot a Swallow, and to bring him to Life again.

Load your gun with the usual charge of powder, but, instead of shot, put in half a charge of quicksilver: prime and shoot. If your piece bears ever so far from the bird, the swallow will find itself stunned to such a degree, as

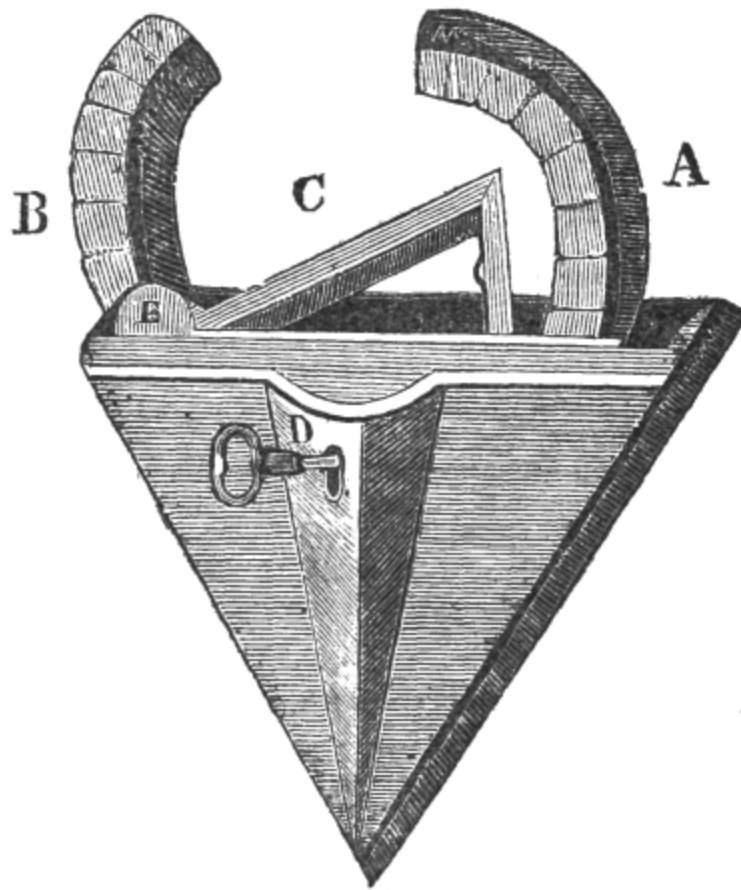
to fall to the ground in a fit. As it will regain its senses in a few minutes, you may make use of the time, by saying that you are going to bring it to life again. This will greatly astonish the company. The ladies will, no doubt, interest themselves in favour of the bird, and intercede for its liberty;—sympathizing with their feelings for the little prisoner may be the means of some of them sympathizing with yours.

Singular Trick with a Fowl.

Take a cock from roost at night, or off its walk by day, and bring him into a room full of company; keep both your hands close to his wings; hold them tight: put him on a table, and point his beak down as straight as possible; and then let any one draw a line with a piece of chalk, directly from his beak, and all the noise you can possibly make, with drums, trumpets, or even crowing of other cocks, will not disturb him from the seeming lethargy, which that position you have laid him in, with the chalked line, has effected. Strange as this seems, yet the certainty of it is past a doubt, as many gentlemen who have, ere this, sported some hundreds on the turf, are ready to assert its truth.

To put a Lock upon a Man's Mouth.

You must have a lock made for this purpose, according to the figure; one side of its bow must be immoveable, as that marked A; the other side is noted B, and must be pinned to the body of the lock, as appears at E.



It must be so pinned that it may play to and fro with ease; this side of the bow must have a leg as at B, and then turn it into the lock; this leg must have two notches filed on the inner side, which must be so managed that one may lock or hold the two sides of the bow as close together as possible, and the other notch to hold the said part of the bow a proportionable distance asunder: so that, being locked upon the cheek, it may neither pinch too hard, nor yet hold it so slight that it may be drawn off. Let there be a key fixed to it to unlock it, as you see at D; and, lastly, let the bow have divers notches filed in it, so that the place of the partition, when the lock is shut home, will the least of all be suspected in the use of the lock. You must now get one to hold a sixpence, edgeways, between his teeth; then take another sixpence, and with your left hand proffer to set it edgeways between a second man's teeth, pretending that your intent is to turn both into which of their mouths they shall desire, by virtue of your words; which he shall no sooner consent to do, but you, by holding the lock privately in

your right hand, with your fore-finger a little down, after certain words, and the lock having hung on a while, may seem to pull the key out of his nose.

To thrust a Bodkin into the Forehead, without Hurt.

Take a hollow bodkin, so that the blade may slip therein as soon as the point is held upwards. Seem to thrust it into your forehead, and, with a little sponge in your hand, bring out blood or wine, making the spectators think that the blood or wine (whereof you may say you have drank very much) runs out of your forehead; then, after showing some appearance of pain, pull away your hand suddenly, holding the point downwards, and it will fall out, and seem never to have been thrust into the haft; but, immediately afterwards, throw that bodkin into your lap or pocket, and pull out another plain one like it, which will completely deceive the spectators.

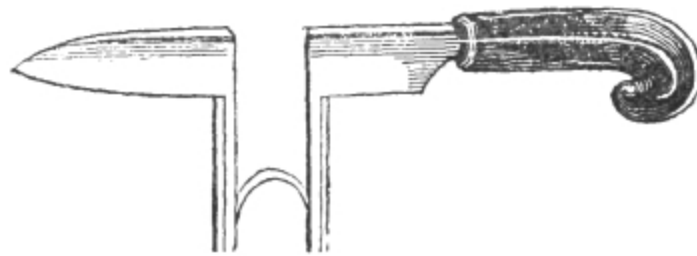
To thrust a Bodkin through your Tongue.

This trick is to be performed in much the same manner as the former, and, as in the first case, without hurt; if the reader will take the pains to practise this trick, he will be in no more danger of performing one than the other.

To render the deception as complete as possible, proceed as follows:— Make a bodkin, or nail, the blade thereof being cut in the middle, so that one part may not be close to the other, by almost three quarters of an inch, each part being kept asunder by a small crooked piece of iron, of the fashion described in the next trick; then thrust your tongue between the aforesaid place, to wit, into the space left in the bodkin-blade, thrusting the said part behind your teeth, and biting it. It will seem to stick so fast in, and through your tongue, that one can hardly pull it out. You must have another bodkin or nail, just like the false one, to show the company.

To appear to cut your Arm off, without Hurt or Danger.

You must provide yourself with two knives, a true one and a false one, and let them be so like each other, that no person can tell one from the other. When you go to show this feat to the company, put the true knife into your pocket; then take out the false one, and clap it on your wrist undiscovered. If, with a sponge, you make the knife bloody, it will seem so much the more strange. This is the form of the knife:—



Tricks with a Cat.

Fix a little bell to the end of a cat's tail, and let her go; she, feeling the tightness of the string, and hearing the bell jingle, will run up and down as if she were mad, flying against the doors and windows; then, if she can, she will get into some hole to hide herself; but, when she wags her tail ever so little, out she comes, and will be as mad as before, and will never rest till it be off. Walnut shells, fastened with a little warm wax or pitch to the cat's feet, will make sport all over the house; and at night, on the stairs, will make the superstitious think that it is a ghost going up and down.

To make a Calf's Head bellow, when served up to Table.

This is effected by the following simple stratagem:—Take a frog that is alive, and put it at the further end of the calf's head, under the tongue, which you must let fall all over it, taking care not to put the frog there till the calf's head is going to be served up. The heat of the tongue will make

the frog croak; which sound, coming from the hollow part of the head, will be like the bellowing of a calf, as if it were alive.

To make a Ball rise above the Water.

Place on the spout of the fountain an inverted cone of wire net-work; throw into it a light hollow copper ball, two inches and a half in diameter, which let fall to the narrow part of the cone to reach the spout: it will rise, and remain suspended in the air till brought down by the wind, and repeatedly re-ascend.

Mode of sealing Letters, whereby an impression cannot be taken.

Take a piece of any aromatic resin, or of amber, and set fire to it on an earthen plate. It will emit a smoke of an agreeable smell: hold your watch-seal over the fumigation; it will become black: then melt some sealing-wax, in the usual way, spread it on paper, and apply to it the seal, prepared as above; and, when you take it off, the engraving will be seen very plainly.

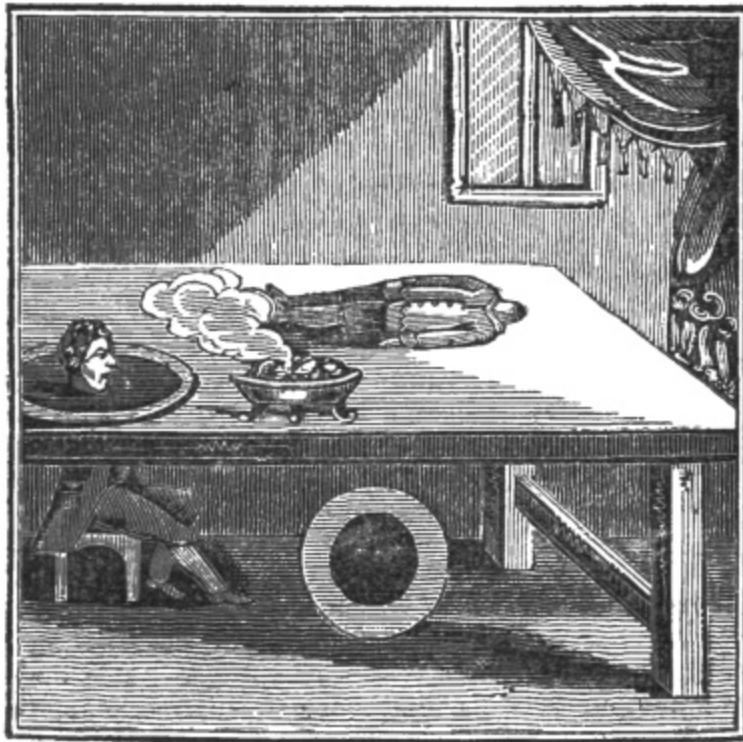
In order to prevent the impression from being taken, flatten the sealing-wax with the blade of a knife: this is easily done, by means of making a thick iron plate red hot, and bringing it close to the impression without touching it; the wax then gets soft, but, before it melts, you must flatten it with a knife, or any other cold instrument. Although thus flattened, the exact impression will remain conspicuous.

The Enchanted Egg.

Have ready a penny-worth of quicksilver in a quill, sealed at both ends with good hard wax; then cause an egg to be boiled, and take off a small bit of the shell of the narrow end, and thrust in your quill of quicksilver, and lay the egg on the ground; you will have sport enough; for it will keep tumbling about as long as there is any heat in it.

To cut a Man's Head off, and to put it into a Platter, a Yard from the Body.

This is a noble trick, if it be performed by a skilful hand. To show it to advantage, you must cause a board, a cloth, and a platter to be purposely made, and in each of them to be made holes, for a man's neck.



The board must be made of two planks, the longer and broader the better; there must be left, within half a yard of the end of each plank, half a hole, so as, both the planks being thrust together, there may remain two holes, like to the holes in a pair of stocks. There must be made likewise a hole in the cloth; a platter, also, must be set directly over or upon one of them, having a hole in the middle thereof, of the like size, and also a piece cut off the same as big as the neck, through which the head may be conveyed into the middle of the platter; and then, sitting or kneeling under the board, let the head only remain upon the board, in the frame. Then, to

make the sight more dreadful, put a saucer, containing saffron, common salt, and spirits of wine, burning, before the head of the boy, who must gasp two or three times. The head will presently appear quite dead, if the boy set his countenance accordingly; and, if a little blood be sprinkled on his face, the sight will be the stranger and more horrible still.

This is practised with a boy instructed for the purpose, who, being familiar with the company, may be known as well by his face as by his apparel. At the other end of the table, where the like hole is made, another boy, of the same size, must be placed, having on his usual apparel; he must lean or lie upon the board, and put his head underneath it, through the said hole; so that the body shall seem to lie on one end of the board, and his head shall lie in a platter on the other end. There are other rules to be observed; as, to put about his neck a little dough, kneaded with bullock's blood, which, being cold, will look like dead flesh, and, being pricked with a sharp quill, filled with blood, will seem to bleed.—You must be particular to have the table-cloth so long and so wide as that it may almost reach the ground. Also, let this be your last trick, taking care not to suffer the company to stay too long in the place after it is performed.

To cause Beer to be wrung out of the Handle of a Knife.

To do this, you must have a small piece of sponge, with liquor put in it privately; then, unseen, place this beyond your right ear; but let not the sponge be too big, or too full of liquor, lest you be discovered. Now, taking a knife, stick it, with the handle upwards, in a table or stool (but observe to place your company before you); then bid them look, saying, "*There you see is nothing of wet, either on this handle or the table;*" so stretch your empty hand towards your ear, darting the point, saying, "*Now, somebody cross my arm,*" and, speaking some powerful words, as "*Jubio bisco,*" then you have a fair opportunity to take the sponge into your hand from behind your ear; stretching forth your hand, squeeze it gently, and afterwards a little harder, which will make it run the faster, to the amazement of the company, at the same time saying, "*Thus could I do till I had drowned you all.*" Then sprinkle a little in their faces, which will cause them to shut their eyes, whilst you convey away the sponge.

To cut a Glass by Heat.

You must have a piece of well-dried match-cord. Having lighted it by the candle, take a beer-glass, and hold the match to the edge of the glass; now have your finger ready wetted, and, when the glass is very hot, clap your finger to the hot place, and it will suddenly crack about a quarter of an inch downward; then keep the coal of the match at the like distance from the end of the crack; and, as it follows, so move your hand, and cut it screw-fashion, otherwise it will not hold together till you have it through to the bottom. When you have done it, and it is cold, take it by the foot, and turn it downwards: it will stretch so that you may put your finger between each cutting; then, when you turn it up again, you may drink a glass of beer from it, and not spill a drop.

THE END.

PRINTED BY G. H. DAVIDSON,
IRELAND YARD, DOCTORS' COMMONS.

Transcriber's Notes

Punctuation and spelling were made consistent when a predominant preference was found in this book; otherwise they were not changed.

Simple typographical errors were corrected; occasional unbalanced quotation marks retained.

Ambiguous hyphens at the ends of lines were retained; occurrences of inconsistent hyphenation have not been changed.

The original mid-paragraph positions of Illustrations have been retained in this eBook, as they often are crucial to understanding the explanations.

The original [Table of Contents](#) used “ib.” for page numbers that were the same as the preceding number, but this eBook uses the actual numbers.

Page [42](#): “then put the” was printed as “then pu- the”. It has been changed to “put” to be consistent with “put out the sixpence” at the end of the same paragraph. Both probably should be “pull”.

Page [61](#): “grove” was printed that way.

Page [68](#): Transcriber added “turn” to “turn the perfect colour of sack wine”.

Page [70](#): “mattrass” was printed that way.

Page [75](#): “gripping” was printed that way.

The bottom of the image of a bottle on page [76](#) was cut-off in the original book.

*** END OF THE PROJECT GUTENBERG EBOOK THE JUGGLER'S
ORACLE; OR, THE WHOLE ART OF LEGERDEMAIN LAID OPEN

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.